Term 7- Sept 2025

Nanoelectronics and Technology (01.119/99.503)-Week 8 Class 2

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06-Nov- 2025

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Outline

- Memory Technology
- Volatile-DRAM, SRAM
    - Working principle
    - Challenges
- Non-volatile Random Access Memory
    - Working principle of Flash memory
    - Challenges

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Electronics: Big Picture

Computation ⟷ Storage

Microprocessors
Microcontrollers
DSP
Other processors

SRAM
DRAM
FLASH

Ref: N. Bhat, CeNSE, IISc Bangalore- Nanoelectronic Device Technology

Memory industry and market

- Forecasts show the global semiconductor memory market rising from around **US $17 billion in 2025** to **US $372.7 billion by 2032**, implying a compound annual growth rate (CAGR) of about 11.7%.
- According to industry reports, DRAM + NAND together form ~25% of the total semiconductor industry revenue.
- Memory is increasingly driving value creation in semiconductors, particularly given data-intensive applications (AI, cloud, IoT, autonomous vehicles) that demand high capacity, high bandwidth memory.

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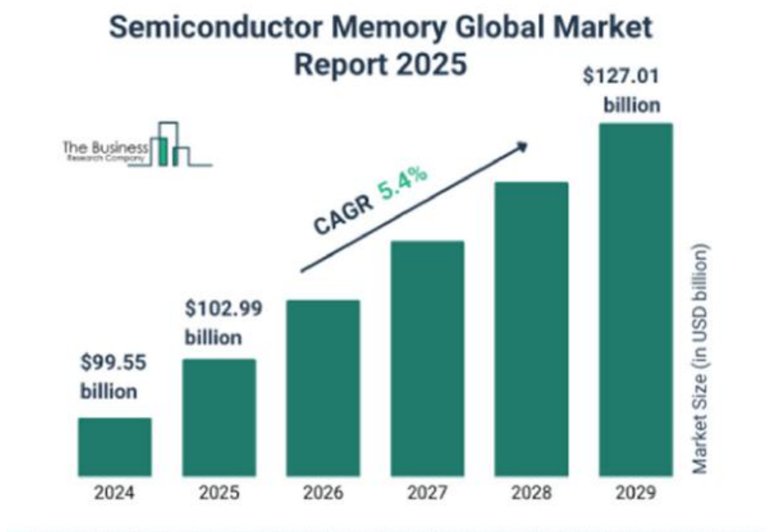
Memory industry and market



Major companies

- Samsung Electronics (South Korea)
- SK Hynix (South Korea)
- Micron Technology (USA)
- Western Digital Corporation / SanDisk (USA)

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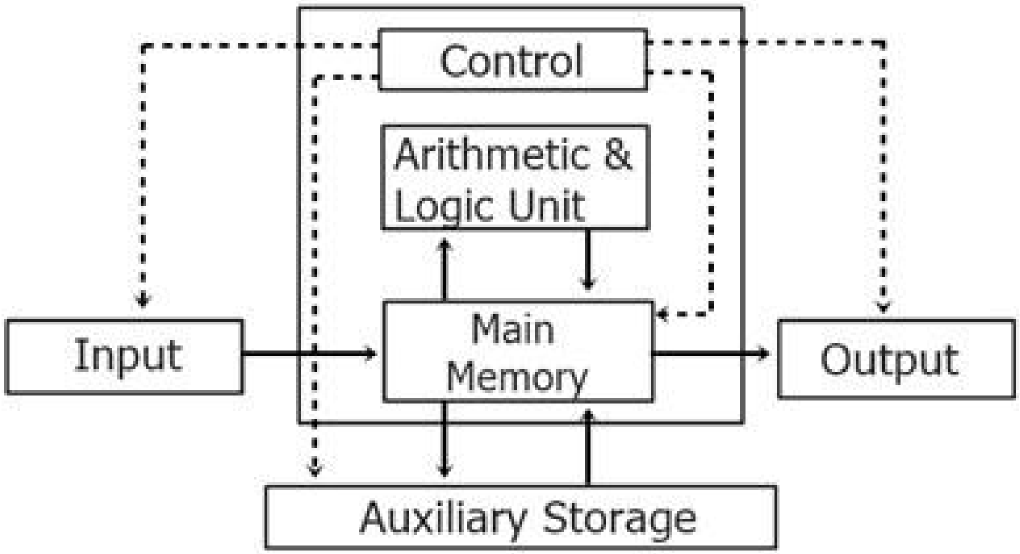
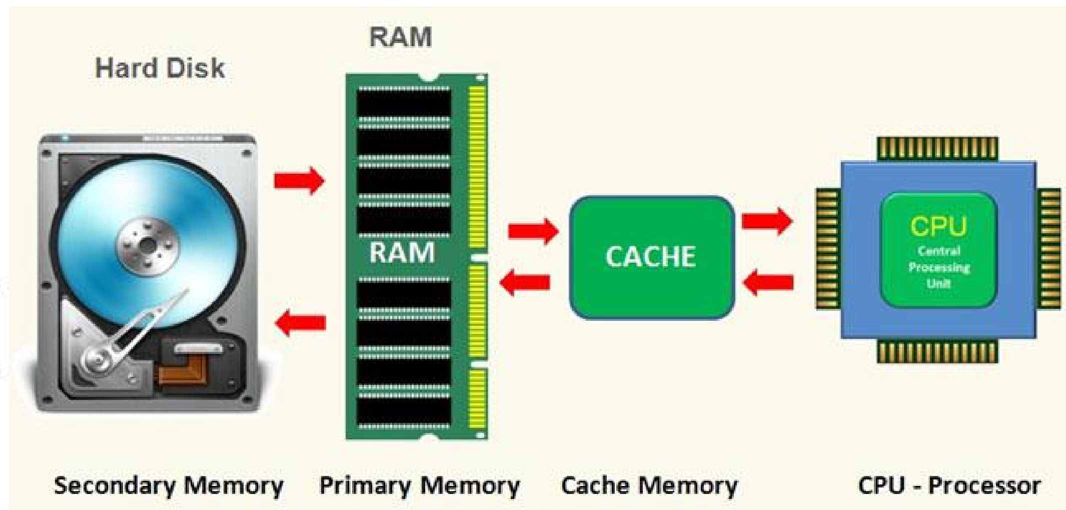
Memory industry and market

https://www.youtube.com/watch?v=R5iHUyNPMtU

Logic and memory





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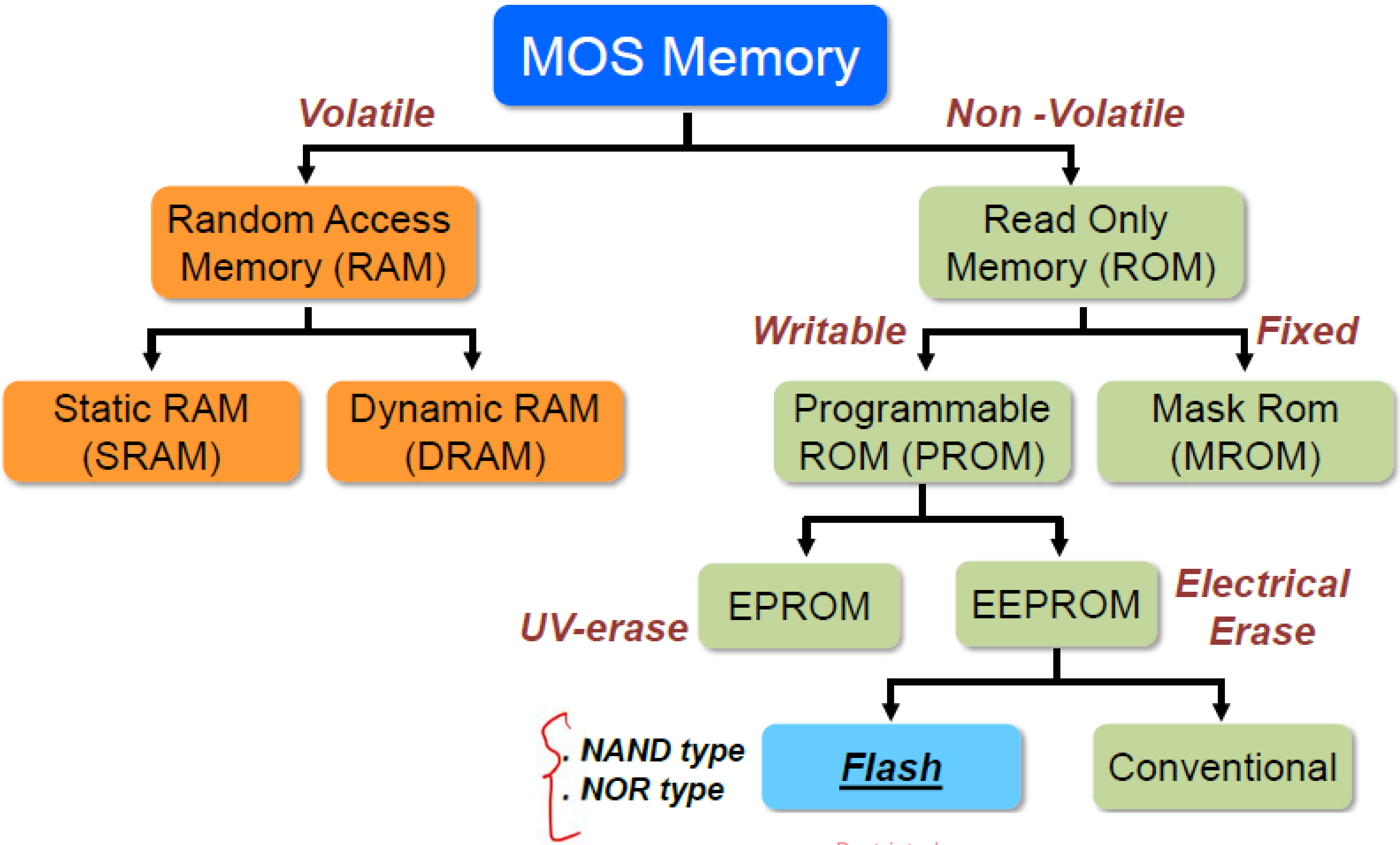
Memory Technologies



J. Kim, "Overview of NAND flash memory", Dept. of CSE NSU

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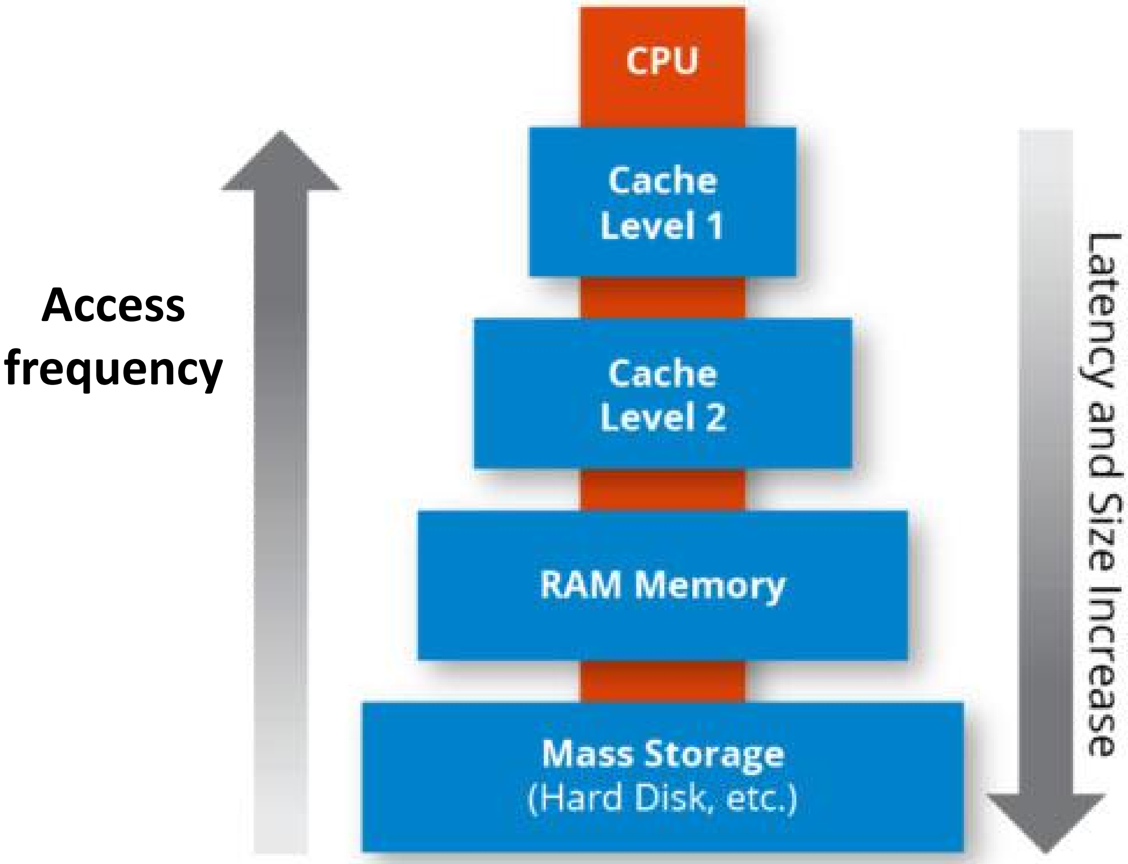
Memory Technologies

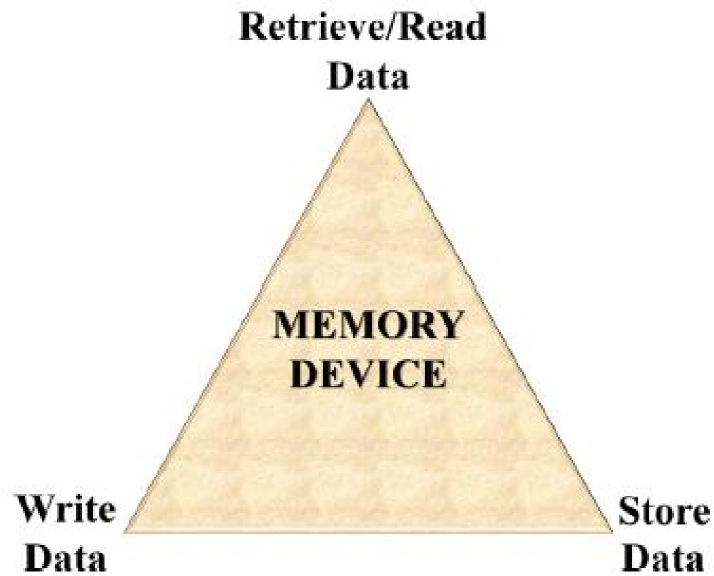
- RAM: Random Access Memory
  - historically defined as memory array with individual bit access
  - refers to memory with both Read and Write capabilities
- ROM: Read Only Memory
  - no capabilities for "online" memory Write operations
  - Write typically requires high voltages or erasing by UV light
- Volatility of Memory
  - volatile memory loses data over time or when power is removed
    - RAM is volatile
  - non-volatile memory stores date even when power is removed
    - ROM is non-volatile
- Static vs. Dynamic Memory
  - Static: holds data as long as power is applied (SRAM)
  - Dynamic: must be refreshed periodically (DRAM)

Prof. A. Mason 'Memory Basics', ECE 410,

Memory Technologies



https://hazelcast.com/glossary/memory-caching/



Three key requirements of a memory device.

S. Bhatti, et al., “Spintronics based RAM”, Materials Today (2017)

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Cache memory is a **small, high-speed memory** located close to the CPU that stores **frequently accessed data and instructions**.

Its purpose is to **bridge the speed gap** between the **fast CPU** and the **slower main memory (RAM)**.

- **Speed:** Much faster than main memory (uses SRAM).
- **Size:** Much smaller (usually a few KB to a few MB).
- **Goal:** Reduce average memory access time by keeping recently used or nearby data close to the processor.

- Main Type: SRAM (Static Random Access Memory)

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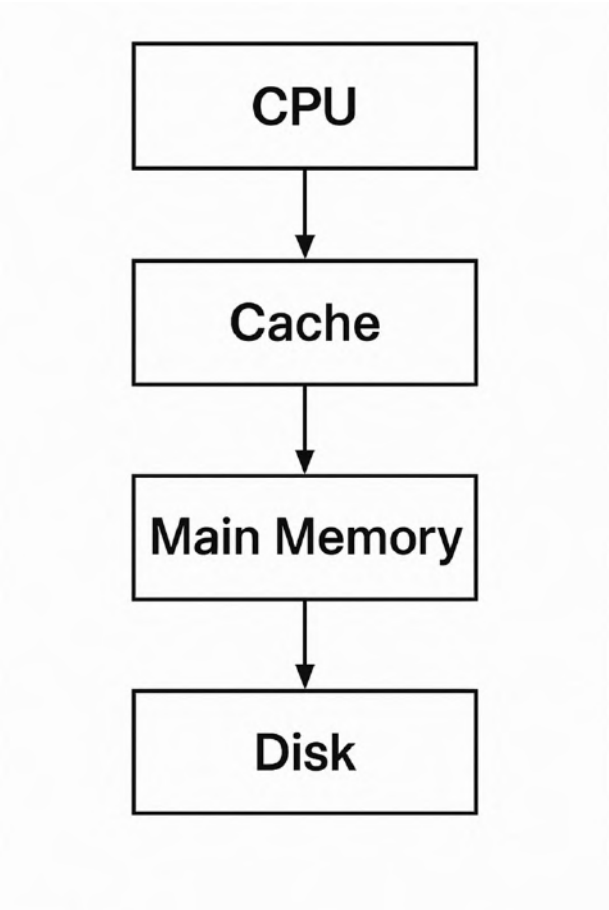
Memory Technologies

Main Memory (RAM)

- **Main memory**, usually called **RAM (Random Access Memory)**, is the **primary storage** the CPU uses to store data and instructions that are currently in use.

- It is made of **DRAM (Dynamic RAM)** cells — each cell stores one bit using a **capacitor and transistor**.



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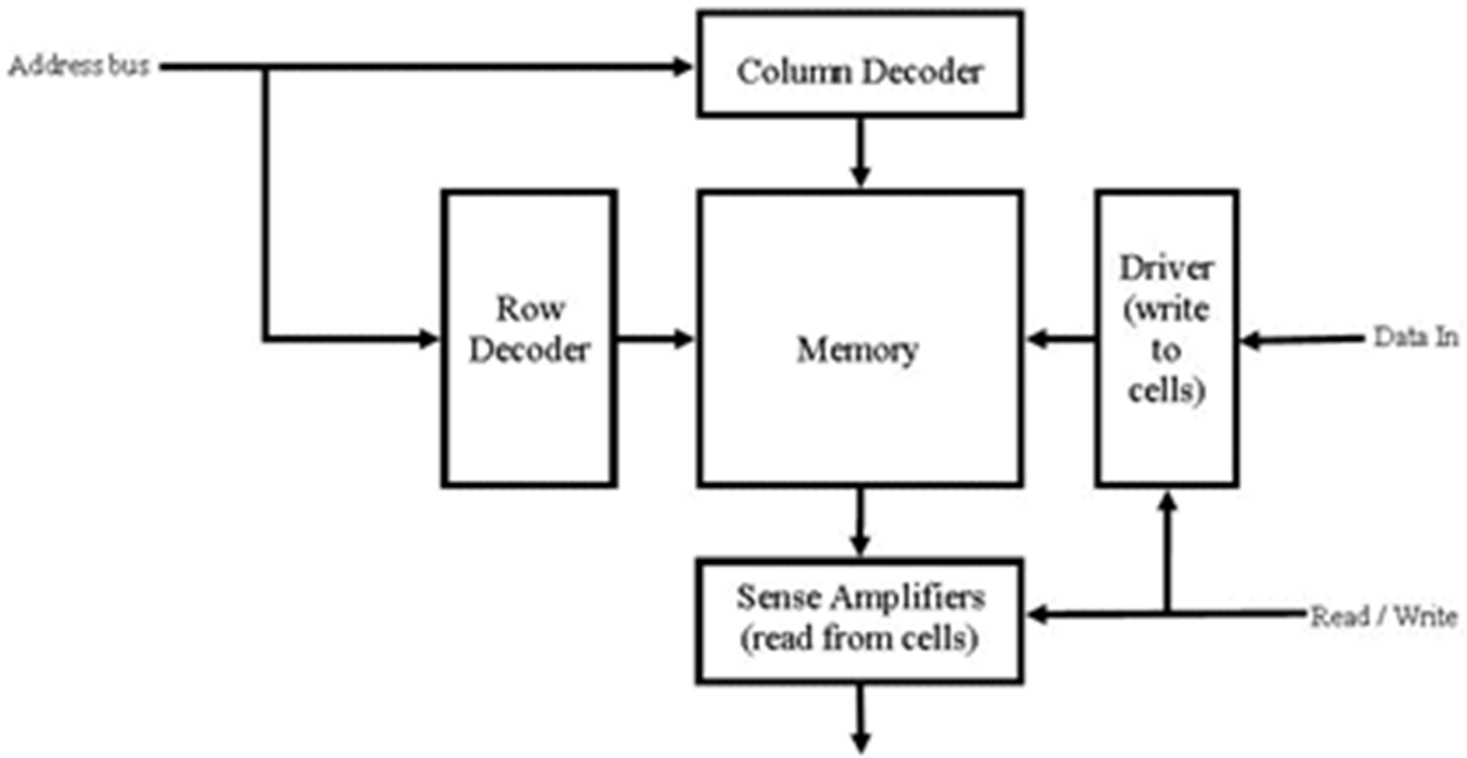
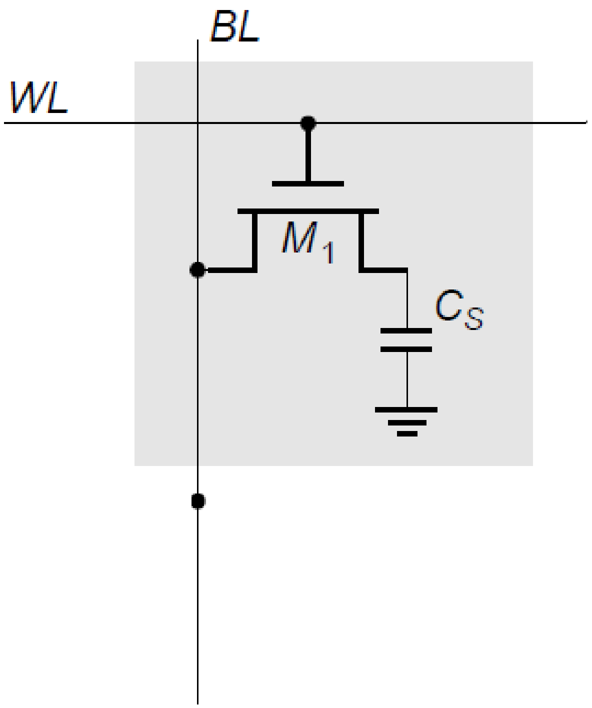
Memory Architecture



DRAM: 1 transistor +1 capacitor

DRAM (Dynamic Random Access Memory) is the most common type of **main memory** used in computers.
It stores **each bit of data in a capacitor** and requires **periodic refreshing** to maintain that data — hence the term *dynamic*.



Each DRAM cell consists of:

1. **One transistor** – acts as a switch to control access to the capacitor.
2. **One capacitor** – stores the charge representing data (`1` or `0`).

- Charged capacitor → logic '1'
- Discharged capacitor → logic '0'

Prof. A. Mason 'Memory Basics', ECE 410,

Ref: MOS memory and technologies-NJIT ECE 271 Dr. Serhiy Levkov

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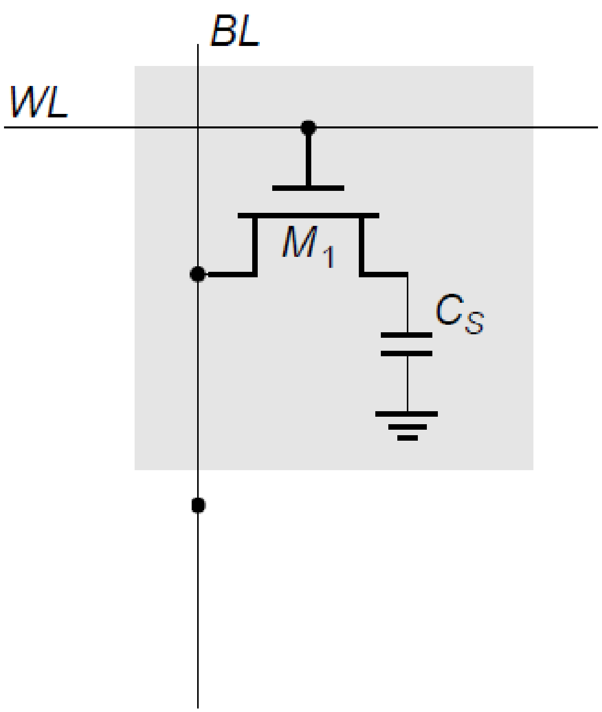

Ref: MOS memory and technologies-NJIT ECE 271 Dr. Serhiy Levkov

- DRAM = Dynamic Random Access Memory
  - Dynamic: must be refreshed periodically
  - Volatile: loses data when power is removed
- Comparison to SRAM
  - DRAM is smaller & less expensive per bit
  - SRAM is faster
  - DRAM requires more peripheral circuitry
- 1T DRAM Cell
  - single access nFET
  - storage capacitor (referenced to VDD or Ground)
  - control input: word line, WL
  - data I/O: bit line

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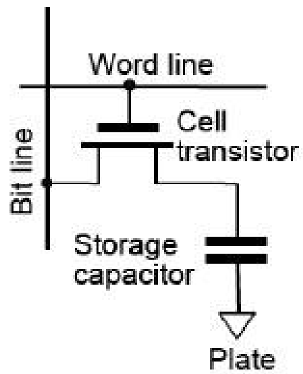
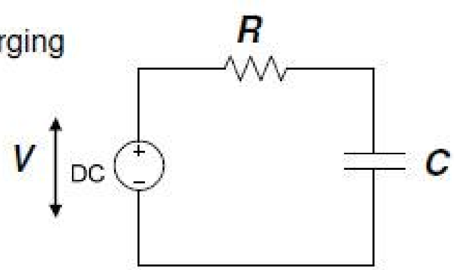
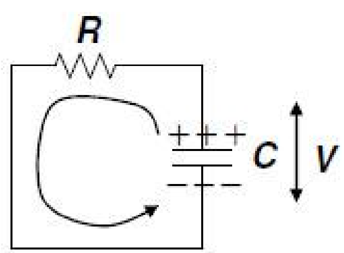
DRAM: 1 transistor +1 capacitor

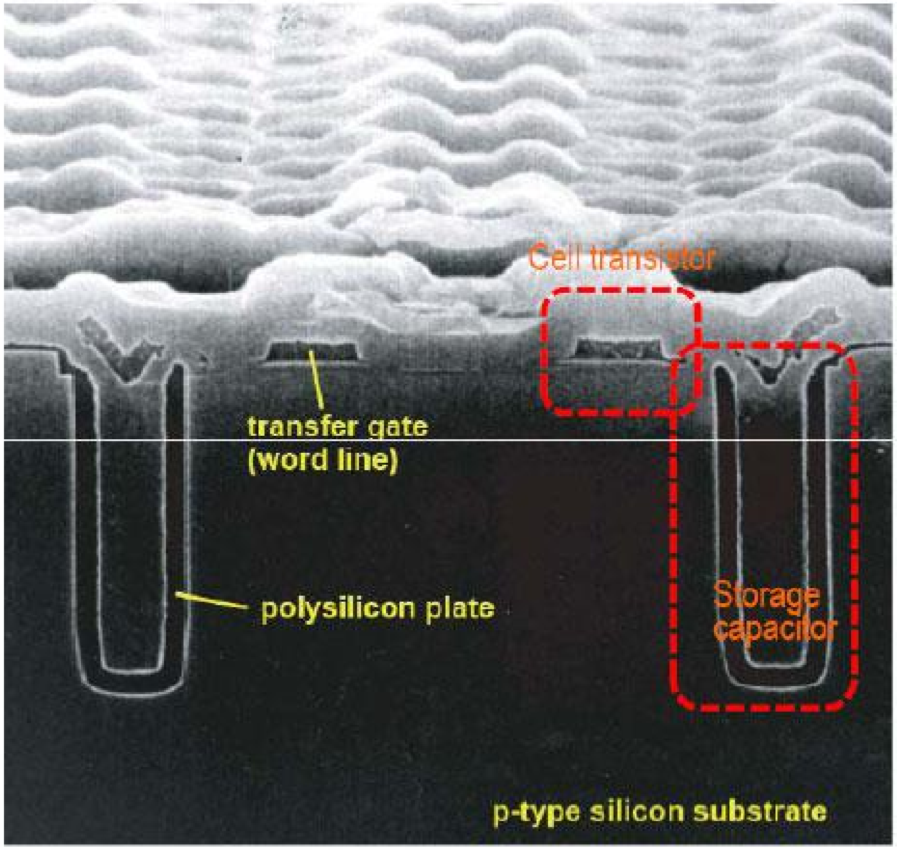


1 charging

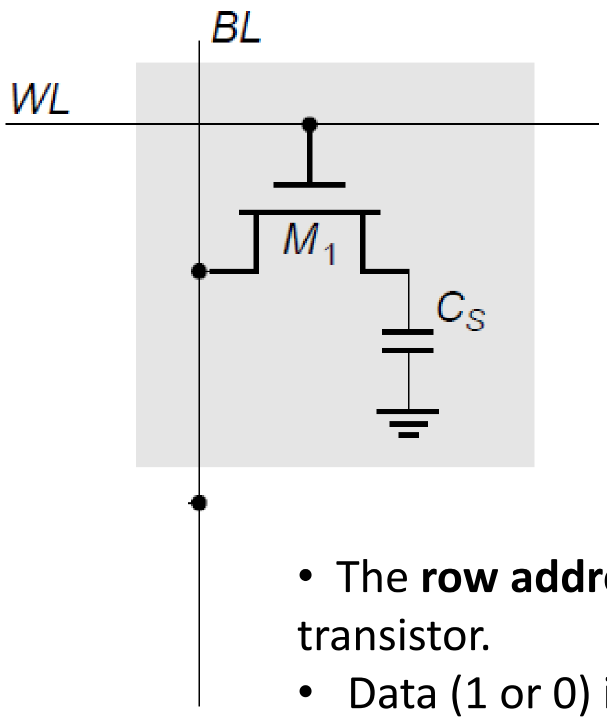


2 leakage





DRAM: Write operation



Drive bitline to Vdd or GND, activate wordline, charge or discharge capacitor

1. Drive bit line
2. Select row

- The **row address** (word line) is activated, turning ON the transistor.
- Data (1 or 0) is sent on the **bit line**.
- Depending on the bit value:
  - If '1': the capacitor is **charged**.
  - If '0': the capacitor is **discharged**.
- The word line is then **deactivated**, trapping the charge.

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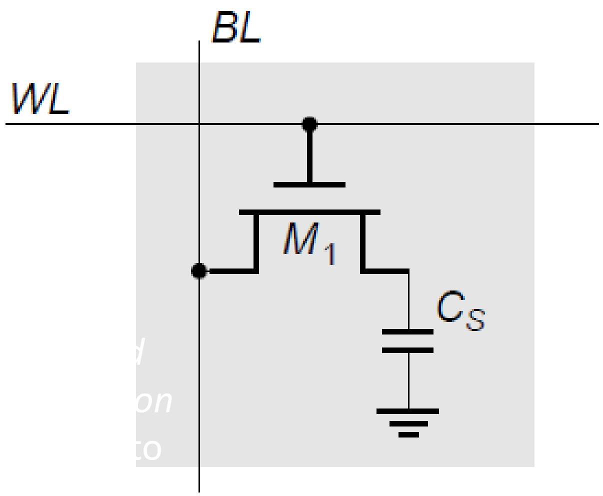
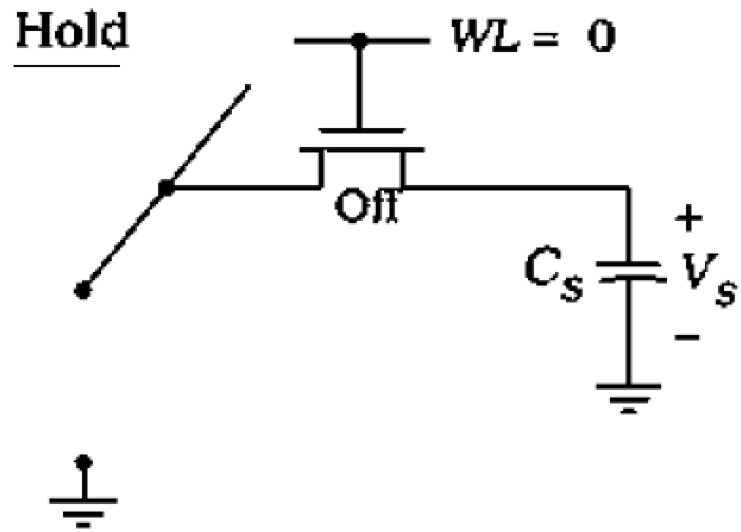
DRAM: Hold operation





Hold operation refers to maintaining the data stored in memory cells. DRAM uses capacitors to store data, with each cell storing a bit as an electrical charge in a capacitor.

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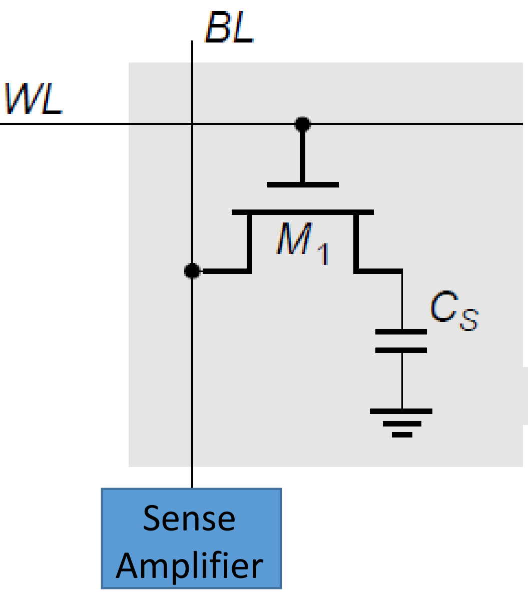
DRAM: Read operation



Precharge bitline to Vdd or Vdd/2

Activate wordline

Capacitor and bitline share charge

- If capacitor was discharged, bitline voltage decreases slightly
- If capacitor was charged, bitline voltage increases slightly

Sense bitline to determine if 0 or 1

- Issue: Reads are destructive! (charge is gone!)
- So, data must be rewritten to cell at end of read

A sense amplifier is a circuit that is used to amplify and detect small signals in electronic systems. It is commonly used in memory circuits, such as dynamic random access memory (DRAM), to read and amplify the weak signals stored in memory cells.

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DRAM: Read operation

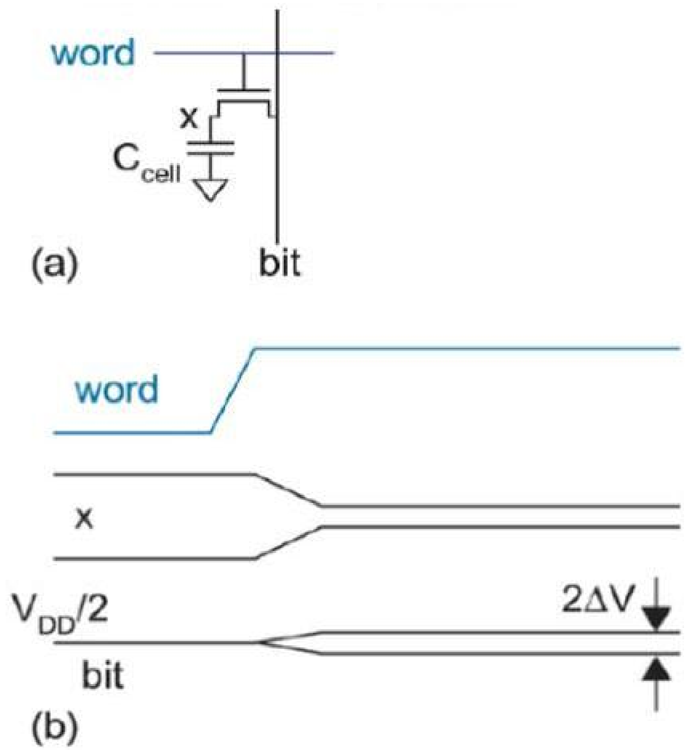
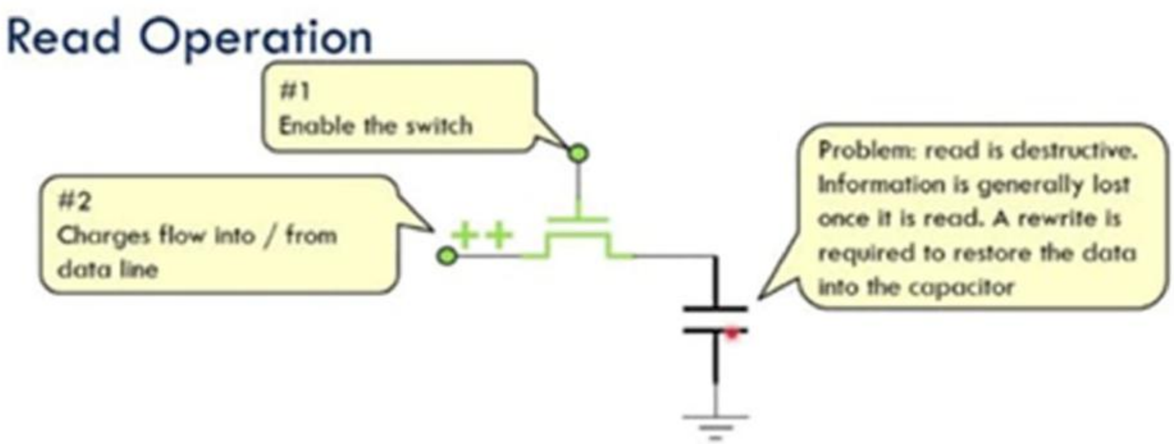


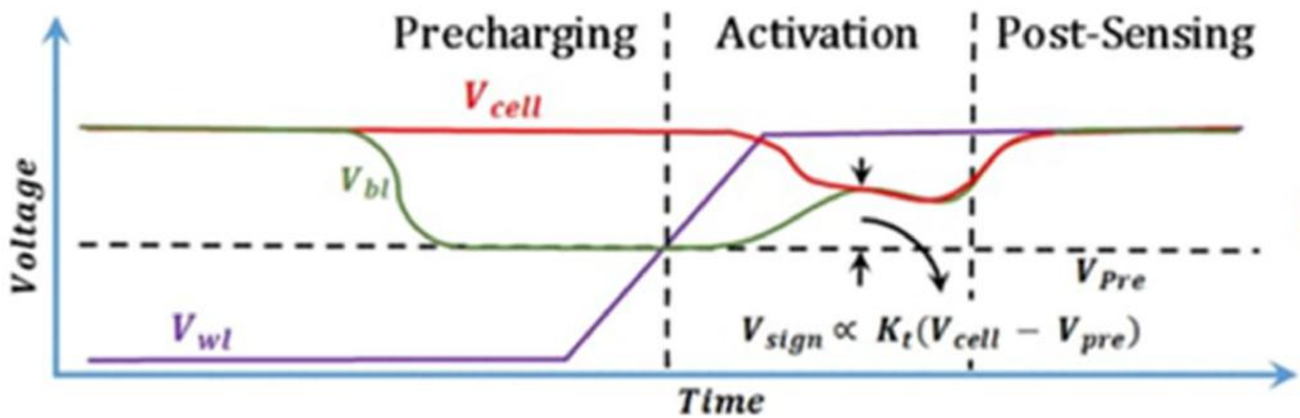
FIG 11.26 DRAM cell read operation

1. bitline precharged to $V_{DD}/2$
2. wordline rises, cap. shares it charge with bitline, causing a voltage ΔV
3. read disturbs the cell content at x, so the cell must be rewritten after each read

$$\Delta V = \frac{V_{DD}}{2} \frac{C_{cell}}{C_{cell} + C_{bit}}$$

DRAM: Read operation





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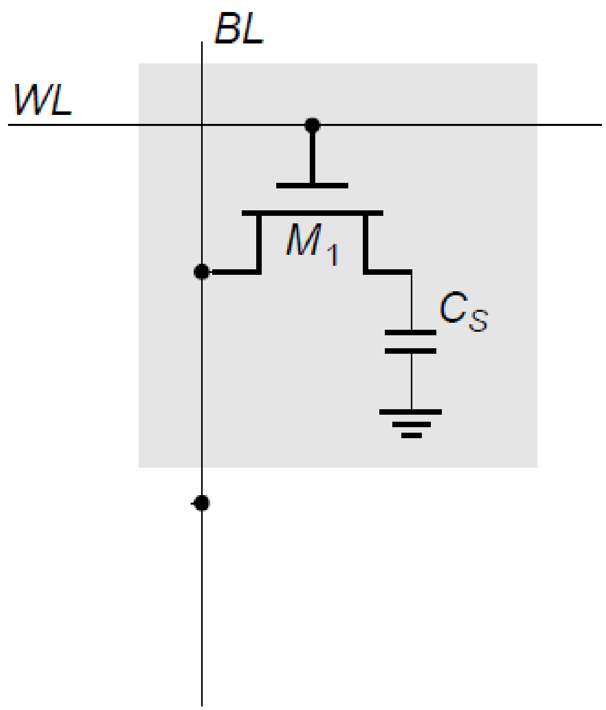
DRAM: Advantages

- High density.
- Smaller region per bit and less expensive.
- Lower power consumption.



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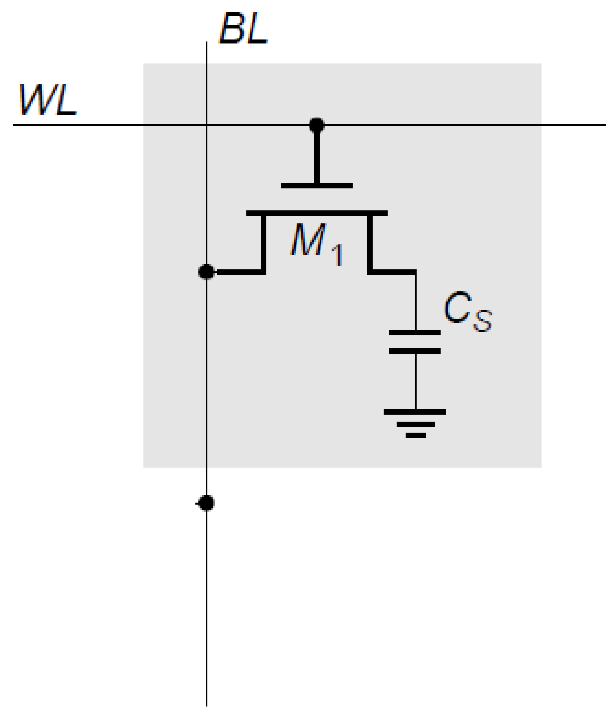
DRAM: Disdvantages

- Needs frequent refreshing due to leakage current
- Slower compared to SRAM



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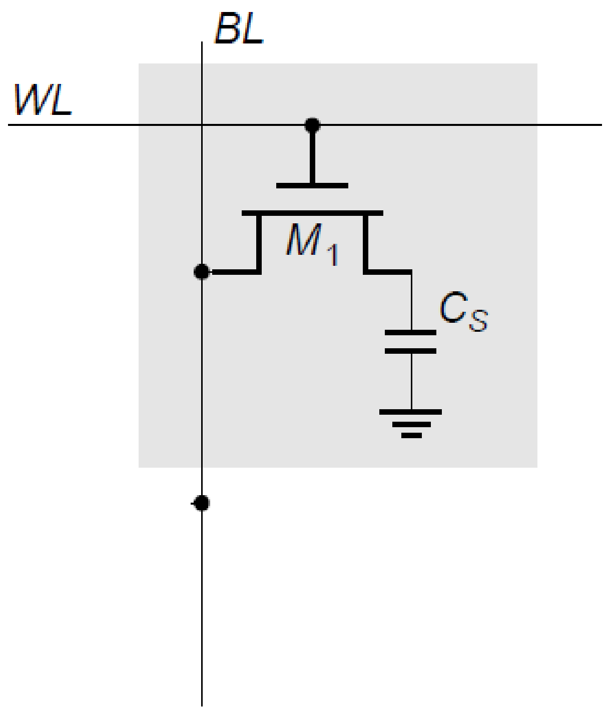
DRAM: Disdvantages



Advantages	**Disadvantages**
High density → large capacity	Slower than SRAM
Low cost per bit	Needs periodic refresh
Compact and simple cell design	Read is destructive (needs rewrite)

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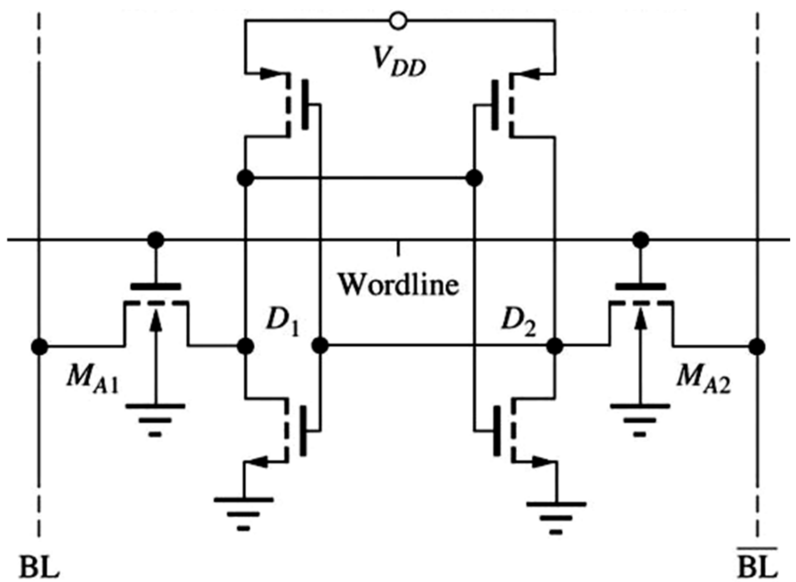
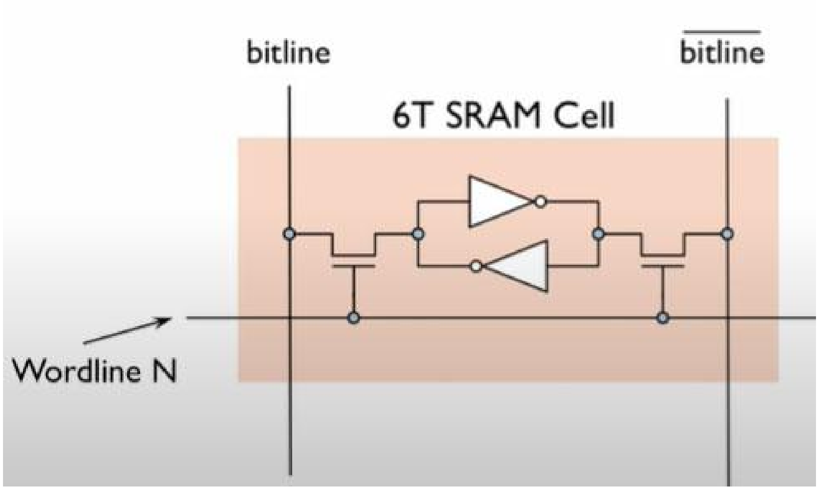
SRAM The 6-T Cell

- **SRAM (Static RAM)** is a type of memory that **stores data using bistable latching circuits (flip-flops)** instead of capacitors (like in DRAM).

- It is **faster and more reliable** but also **more expensive** and **less dense**.

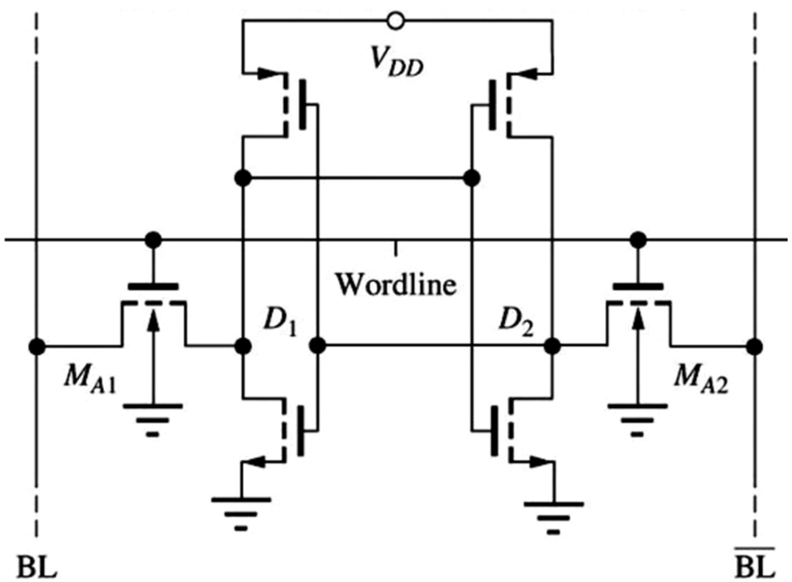


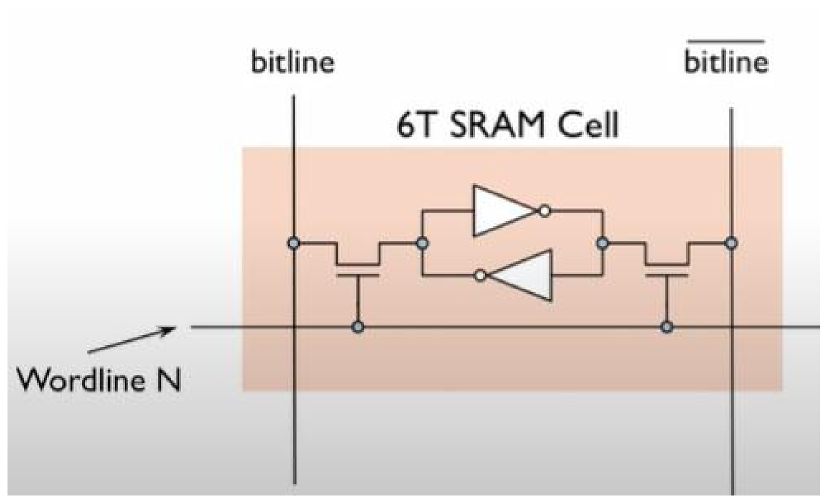


Ref: MOS memory and technologies-NJIT ECE 271 Dr. Serhiy Levkov

SRAM The 6-T Cell

- The previous cell can exist indefinitely long in one of the stable states – high or low, i.e. it can store one bit of information – 0 or 1.
- This is accomplished by addition of two control (access) transistors → a called 6-T (6 transistor) cell that can store 0 and 1values of the data and allows to “write” and to “read” that data.
- The access transistors also isolate one cell from another in a memory array.





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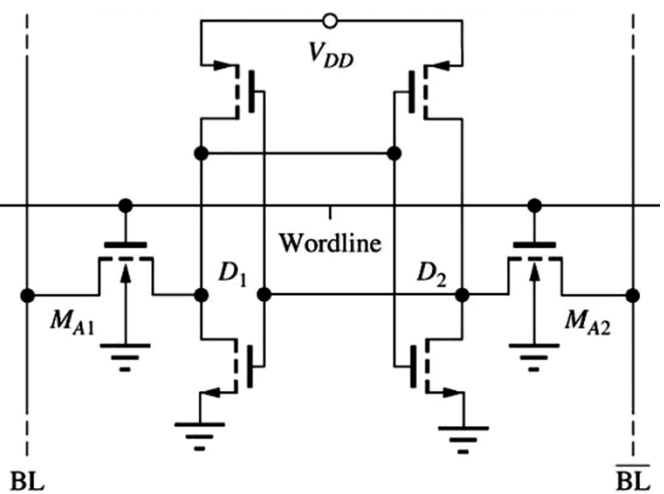

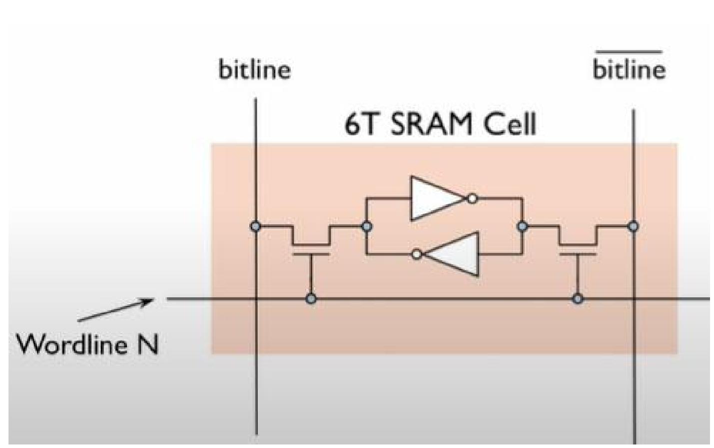

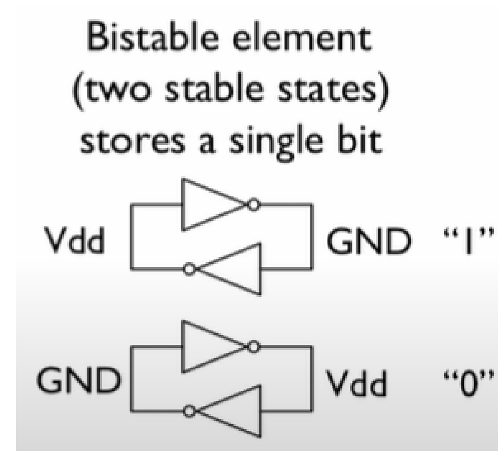

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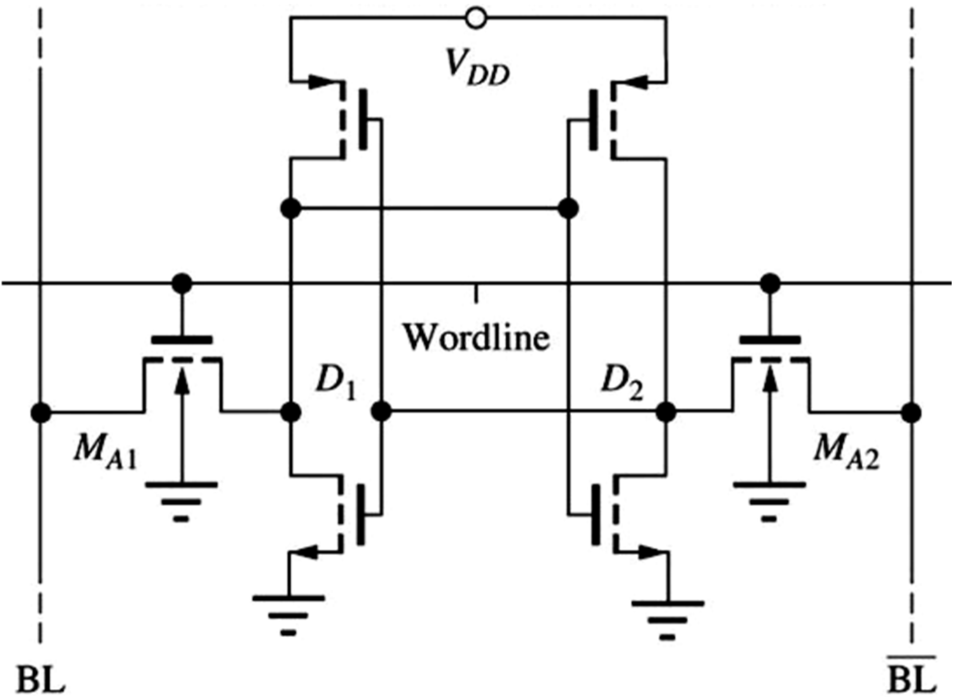
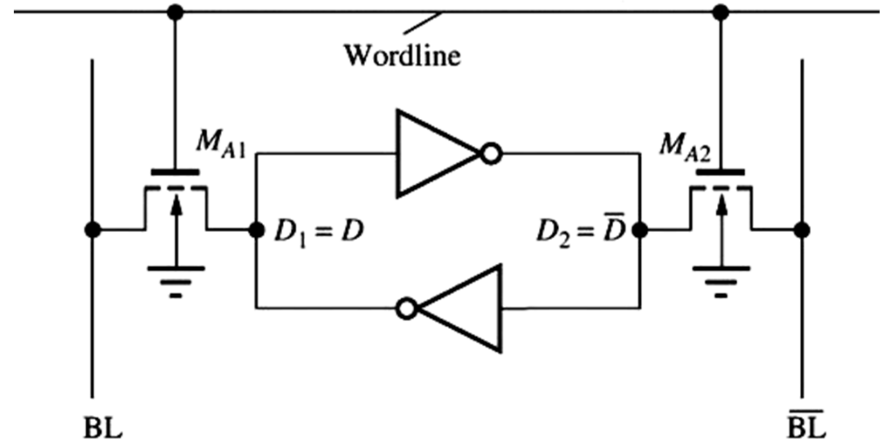
SRAM: 6-T Cell

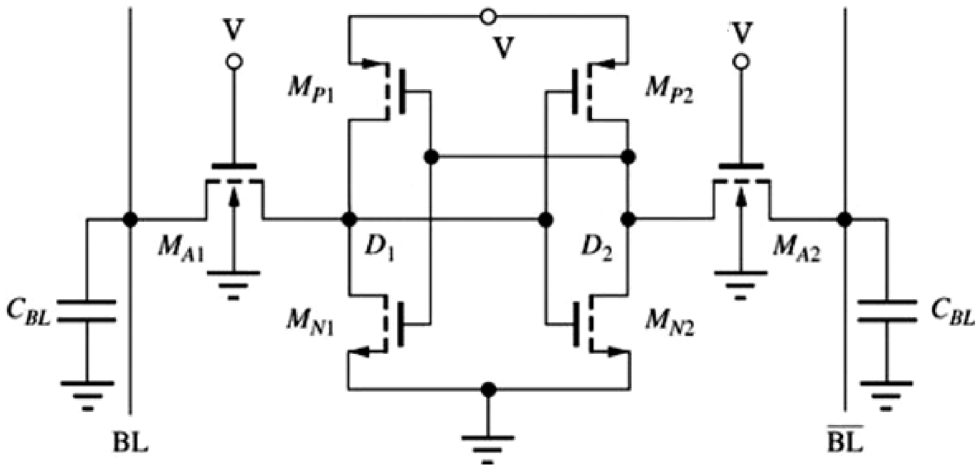


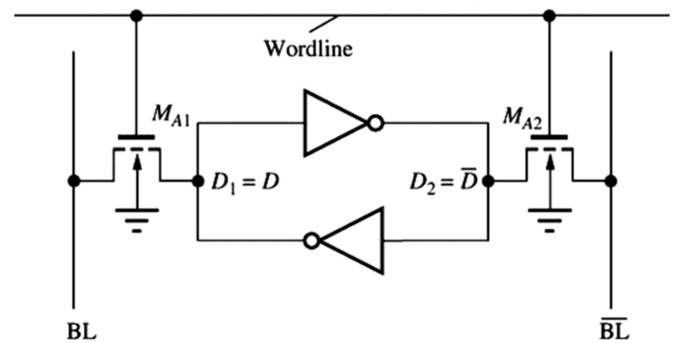


- **NMOS access transistors**
- **Read and write uses the same port: need sufficient margins**
- **One wordline to access cell**
- **Two bit lines (BL, BLB) to carry the data**

Ref: MOS memory and technologies-NJIT ECE 271 Dr. Serhiy Levkov

SRAM: 6-T Cell: Write operation





Write

- word line = 1, access tx are ON
- new data (voltage) applied to bit and bit_bar
- data in latch overwritten with new value

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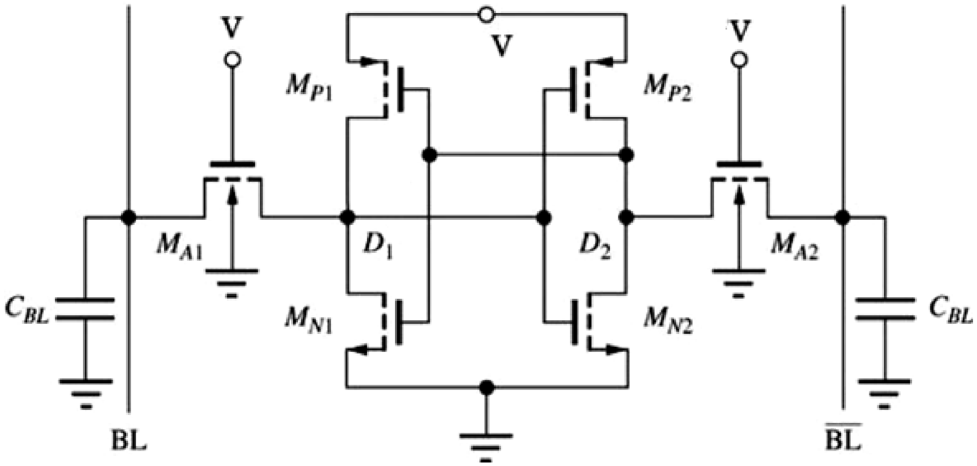
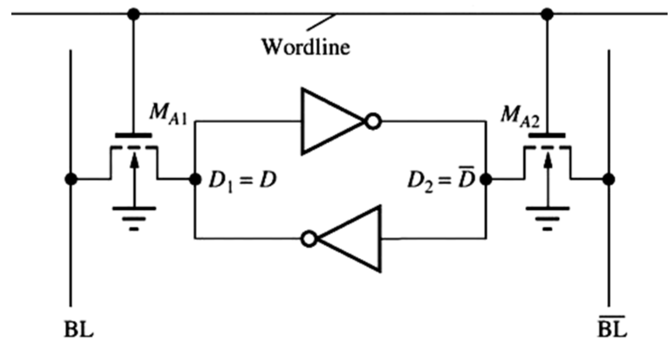
SRAM: 6-T Cell: Read operation





Read

- word line = 1, access tx are ON
- bit and bit_bar read by a sense amplifier

Sense Amplifier

- basically a simple differential amplifier
- comparing the difference between bit and bit_bar
  - if bit > bit_bar, output is 1
  - if bit < bit_bar, output is 0

Ref: MOS memory and technologies-NJIT
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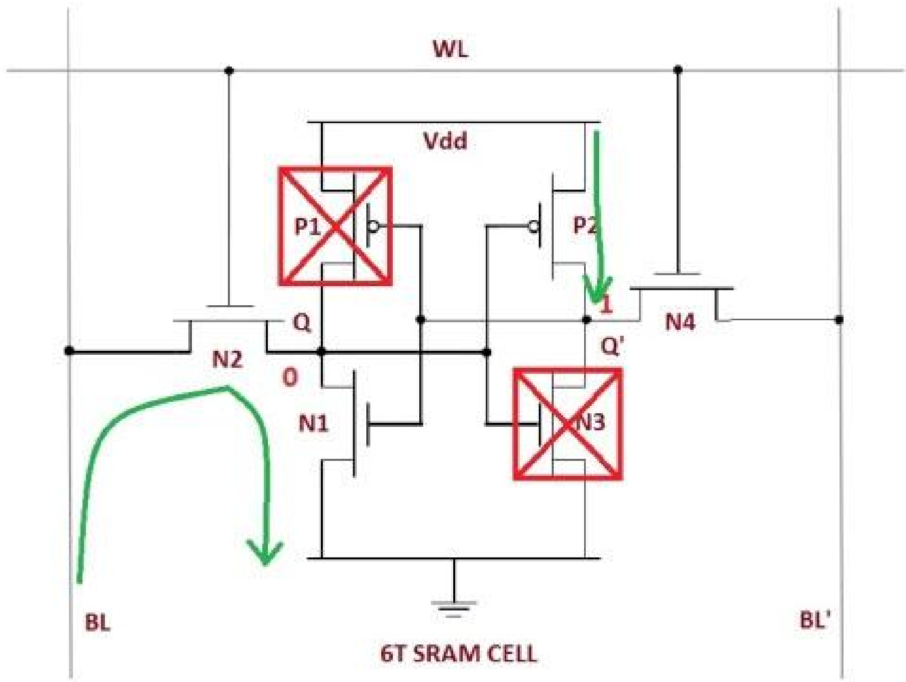
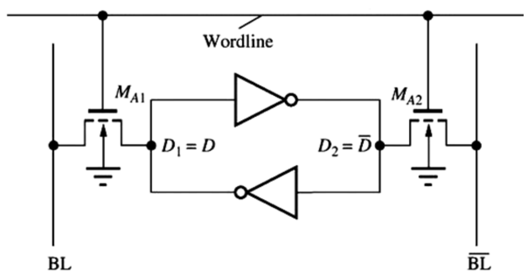
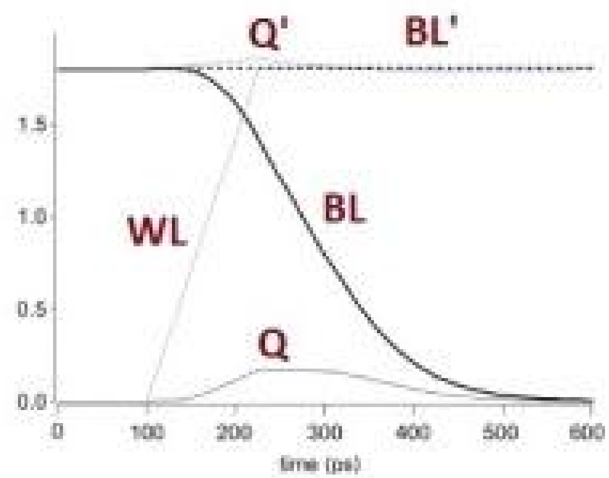
SRAM: 6-T Cell: Read operation



READ OPERATION: i) Precharge BL, BL' to HIGH
ii) Turn on WL
iii) BL or BL' will be pulled down to LOW depending on Q, Q'
iv) Eg. If Q = 0, Q' = 1, BL discharges through N2 - N1 - GND and BL' stays high. But Q bumps up slightly
v) In order for Q to not flip N1 should be stronger than N2, i.e N1 >> N2





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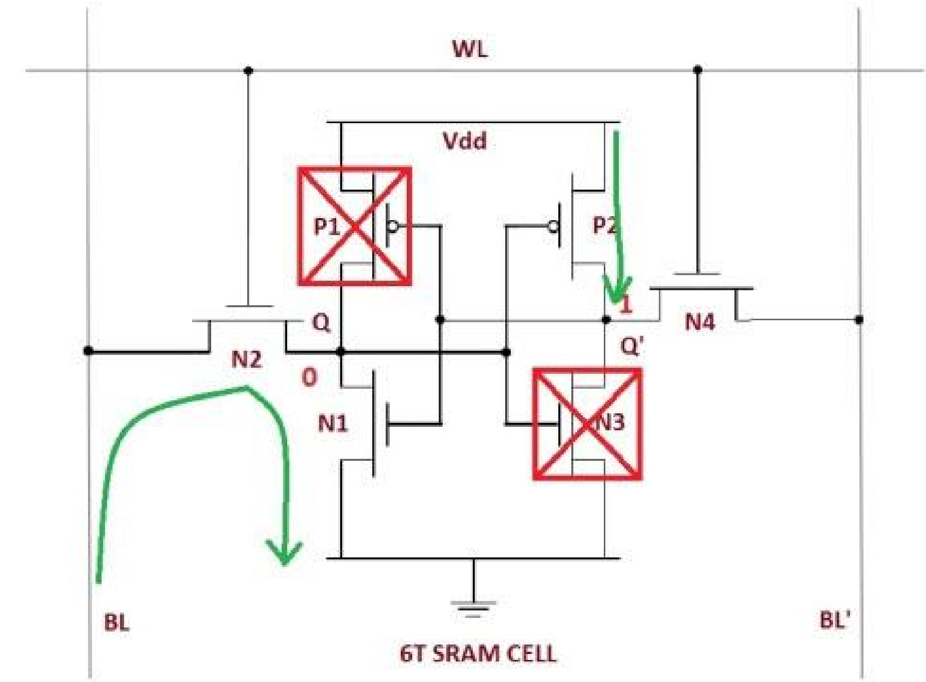
SRAM: 6-T Cell: Read operation



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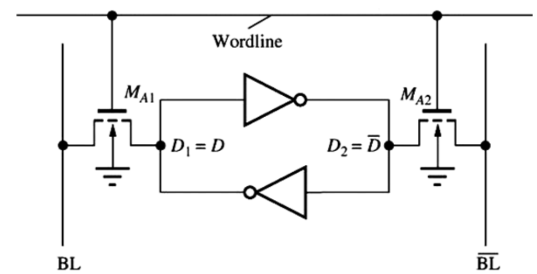

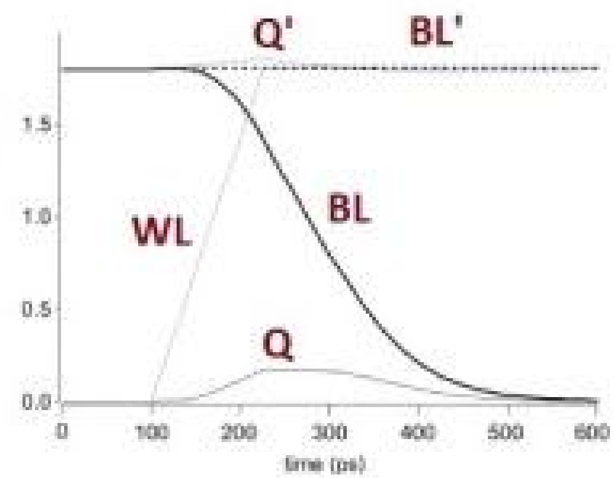

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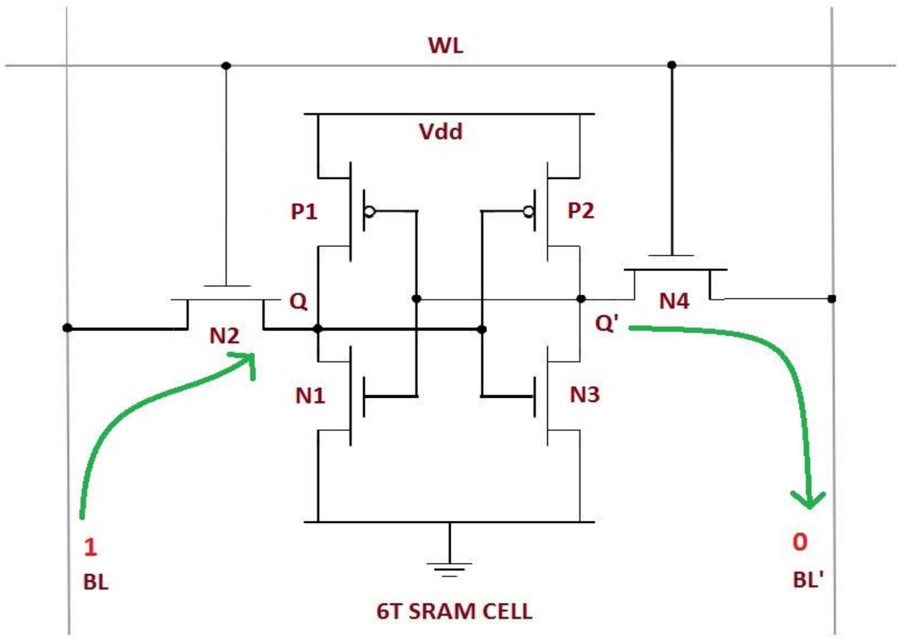
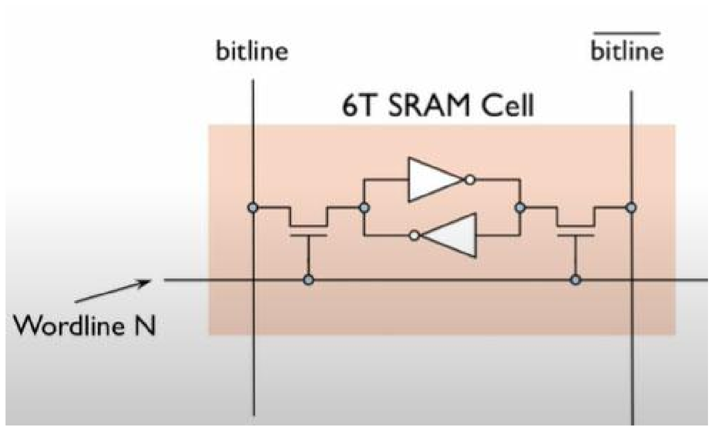
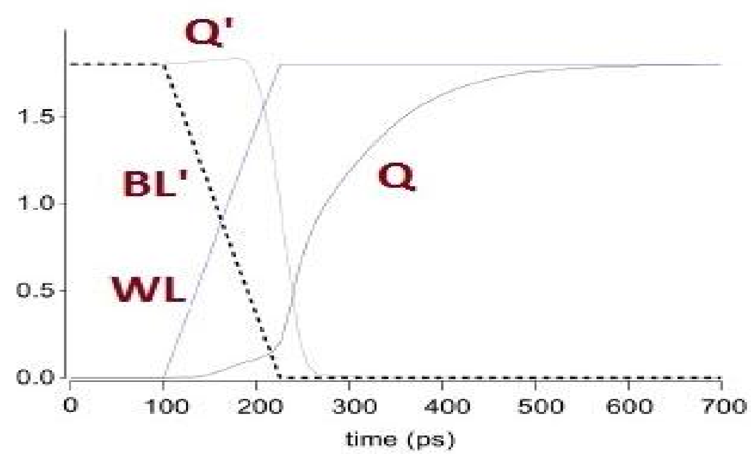
SRAM: 6-T Cell: Write operation

To write "0"







WRITE OPERATION: i) Drive BL, BL' with necessary values
ii) Turn on WL
iii) Bit Lines overpower cell with new value
iv) Eg. Q = 0, Q' = 1 and BL = 1, BL' = 0. This forces Q' to low and Q to high
v) To overpower feedback inverter loop, N2 should be stronger than P1, i.e N2 >> P1

As Q starts to charge up to 1, output of inverter P2-N3 starts to discharge which in turn, makes P1 turn on. Thus the feedback inverters lock on to the values to be written

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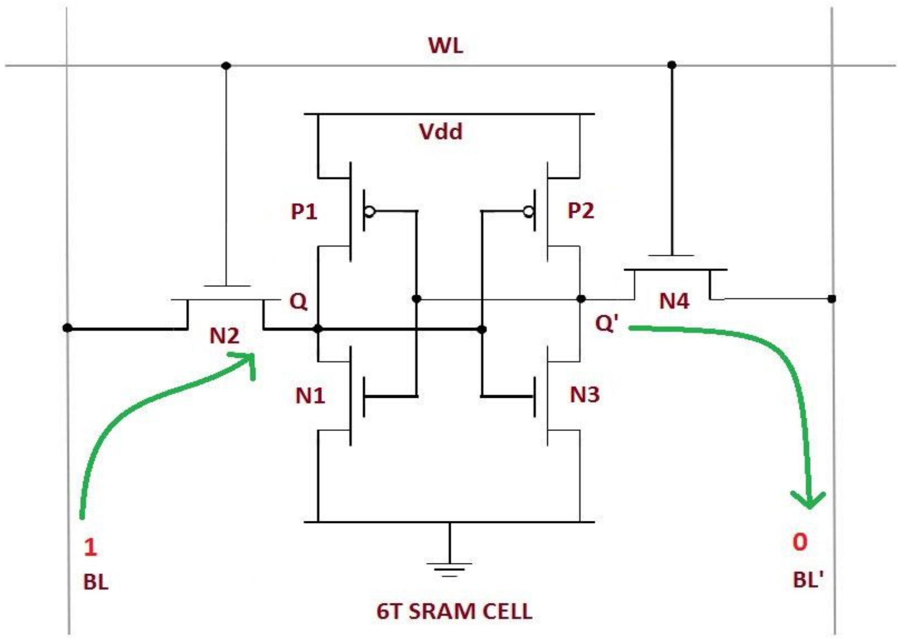
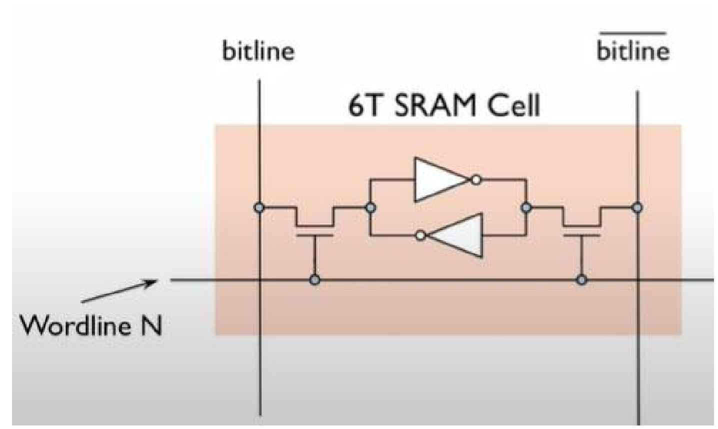
SRAM: 6-T Cell: Write operation

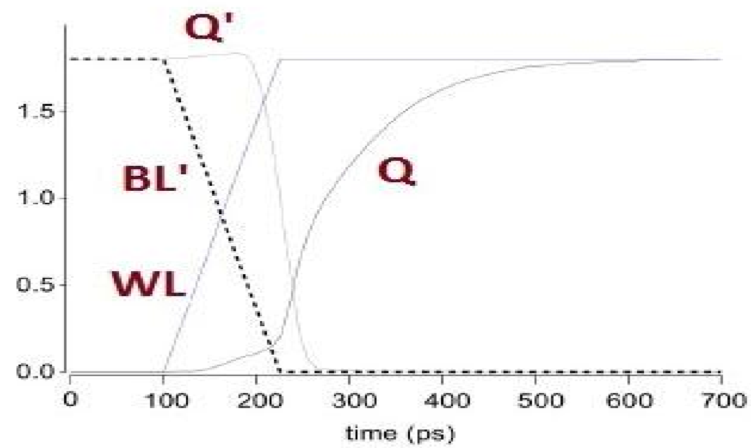
To write "1"



WRITE OPERATION: i) Drive BL, BL' with necessary values
ii) Turn on WL
iii) Bit Lines overpower cell with new value
iv) Eg. Q = 0, Q' = 1 and BL = 1, BL' = 0. This forces Q' to low and Q to high
v) To overpower feedback inverter loop, N2 should be stronger than P1, i.e N2 >> P1

As Q starts to charge up to 1, output of inverter P2-N3 starts to discharge which in turn, makes P1 turn on. Thus the feedback inverters lock on to the values to be written





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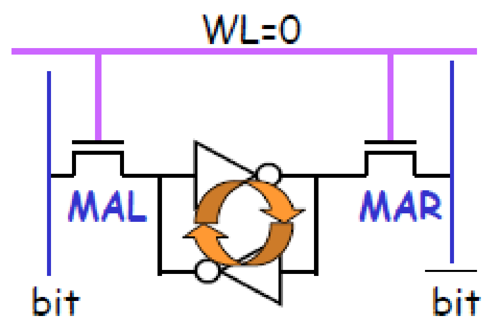
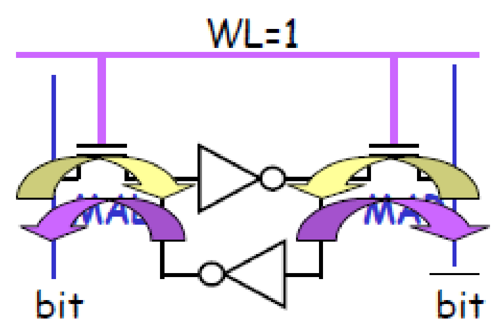
SRAM The 6-T Cell

- Hold
  - word line = 0, access transistors are OFF
  - data held in latch
- Write
  - word line = 1, access tx are ON
  - new data (voltage) applied to bit and bit_bar
  - data in latch overwritten with new value
- Read
  - word line = 1, access tx are ON
  - bit and bit_bar read by a sense amplifier
- Sense Amplifier
  - basically a simple differential amplifier
  - comparing the difference between bit and bit_bar
    - if bit > bit_bar, output is 1
    - if bit < bit_bar, output is 0
    - allows output to be set quickly without fully charging/discharging bit line





Prof. A. Mason 'Memory Basics', ECE 410,

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SRAM: 6-T Cell: Advantages

- SRAM is faster than DRAM which means it is faster in operation.
- SRAM can be used to create a speed-sensitive cache.
- SRAM only has medium power consumption.
- SRAM has a shorter cycle time since it does not require pausing between accesses. No refreshing required

SRAM: 6-T Cell: Disadvantages

- Complex circuit and requires larger silicon area for fabricating single cell.
- High power consumption and heat generation.
- Low density
- Higher cost.

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SRAM: 6-T Cell: Significance

- **SRAM- Cache Memory -**is a technique in which computer applications temporarily store data in a computer's main memory to enable fast retrievals of that data.
- Since accessing RAM is significantly faster than accessing other media like hard disk drives or networks, caching helps applications run faster due to faster access to data.
- By storing the calculations in a cache, the system saves time by avoiding the repetition of the calculation.
- Since the cache is limited in size, eventually some data already in the cache will have to be removed to make room for new data that the application most recently accessed.

SRAM: 6-T Cell: Significance

Parameter	**Typical Range**	**Notes**
Access Time	1–5 ns	Much faster than DRAM
Power	Higher (active)	Consumes static power
Cell Area	~120–150 F^2	Larger than DRAM (~6–8 F^2)
Density	Lower	4–6× fewer bits per area
Data Retention	As long as powered	No refresh needed

Memory Technologies- SRAM and DRAM

Feature	**Cache Memory**	**Main Memory (RAM)**
Type	SRAM (Static RAM)	DRAM (Dynamic RAM)
Access Time	1–5 nanoseconds	50–100 nanoseconds
Location	Inside or very close to CPU	On motherboard (farther from CPU)
Storage Cell	6 transistors (no refresh needed)	1 transistor + 1 capacitor (needs periodic refresh)
Speed	Very fast	Slower
Cost per bit	High	Lower
Size	Small (few MB)	Larger (several GB)

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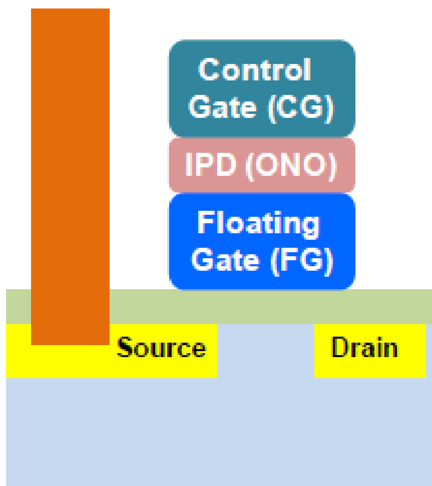
Flash memory-Nonvolatile

- Electrons can be injected (ejected) into (out of) the FG through tunnel oxide with electric field across tunnel oxide
- Cell Vth changes depending on the amount of charge in floating gate (FG)



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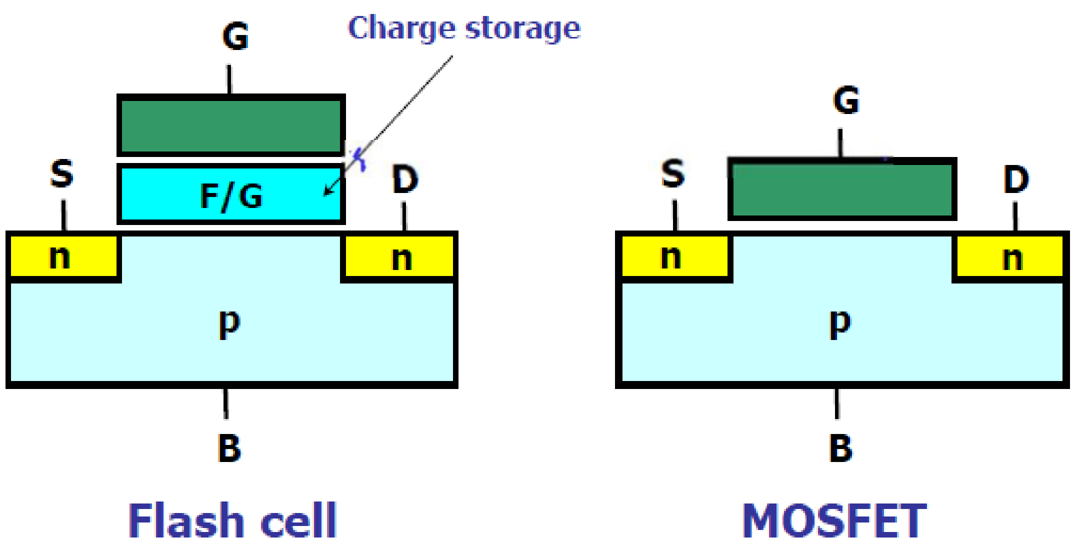
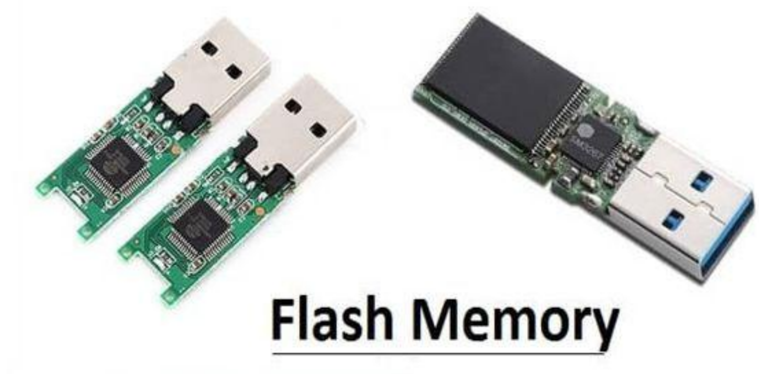
Flash memory: Non-volatile

Flash memory cell vs. MOSFET

- **Flash cell has a charge storage layer such that Vth of a cell can be changed → memorize information**





K. Choi, Samsung Electronics, co., Ltd
Flash design team 'Nand flash memory' 2010.

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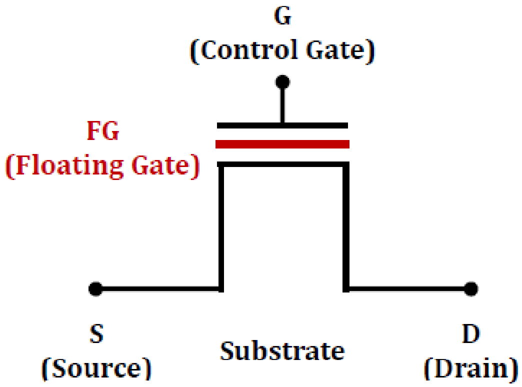
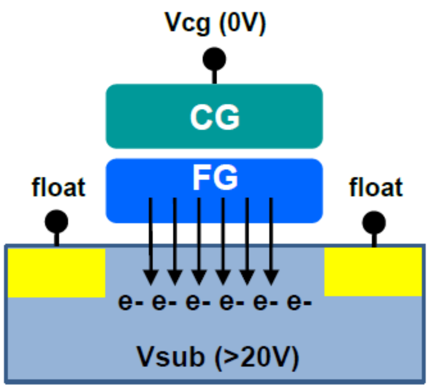
Flash memory: Non-volatile

Flash memory cell

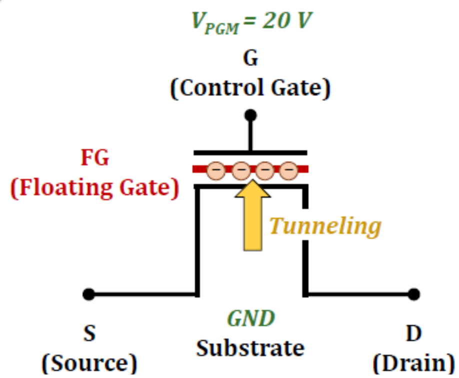


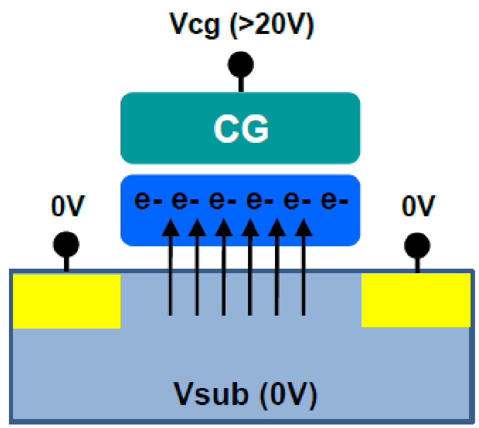


Erase : FN tunneling

On cell → Data “1”

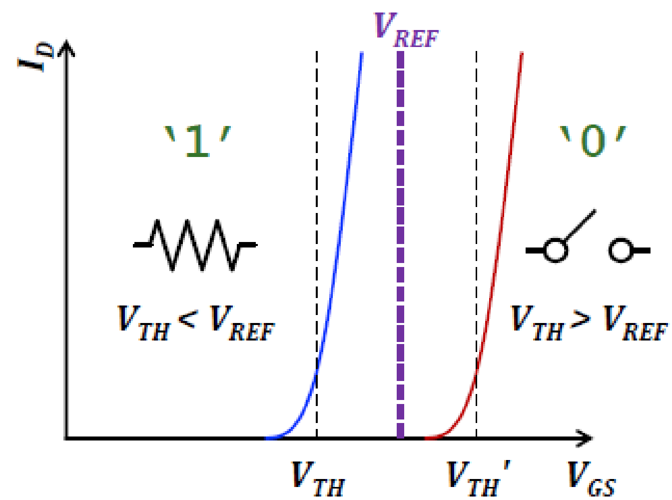
Jihong Kim (SNU) NAND Flash Overview.





Program : FN tunneling

Off cell → Data “0”



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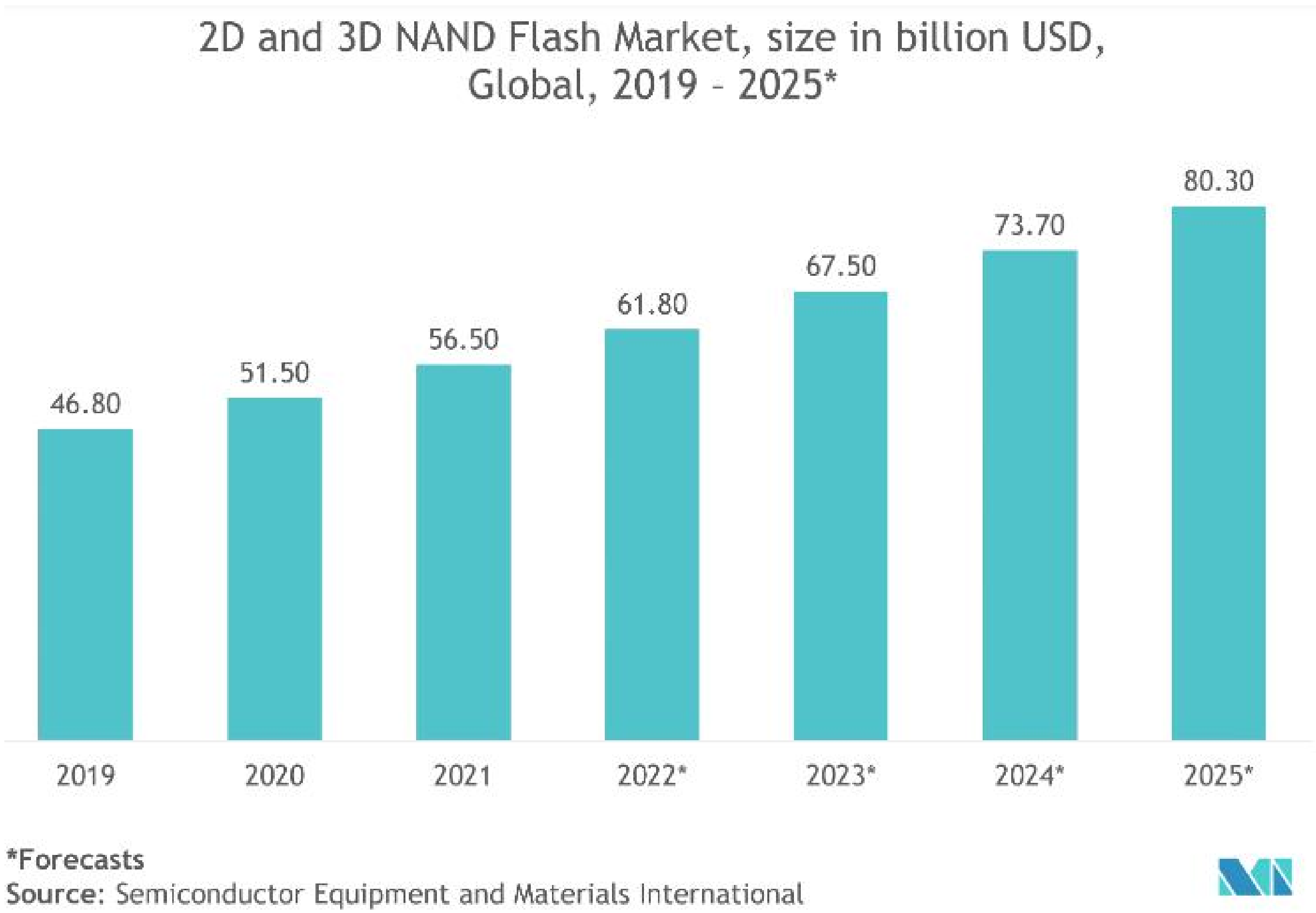
Flash memory market:



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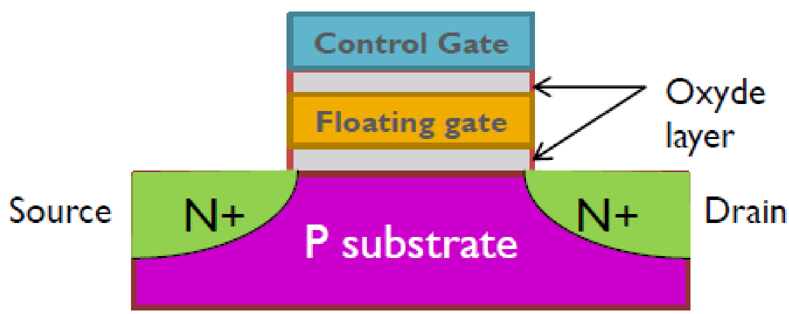
Flash memory: Non-volatile

- Invented by F. Masuoka Toshiba 1980
- Introduced by Intel in 1988
- Type of EEPROM (Electrically Erasable & Programmable Read Only Memory)
- Use of Floating gate transistors
- Electrons pushed in the floating gate are trapped



J. Boukhobza, S. Rubini, Lab-STICC, Université de Bretagne Occidentale, 'Overview of Flash memory'

- 3 operations: program (write), erase, and read

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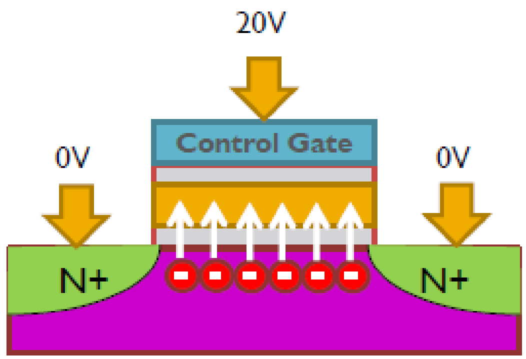
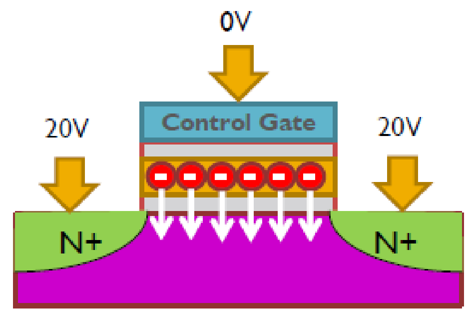
Flash memory: Non-volatile



Program / write operation

- Apply high voltage to the control gate
- → electrons get trapped into the floating gate

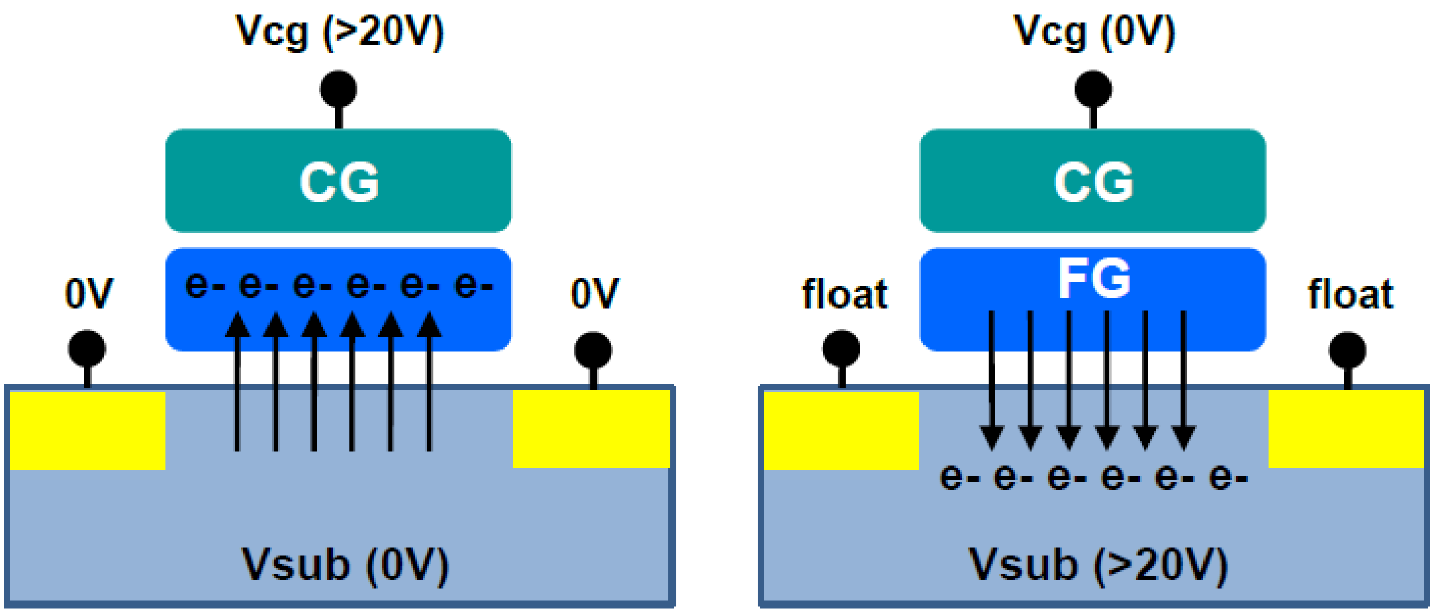


Erase operation

- FN (Fowler-Nordheim) tunneling: Apply high voltage to substrate (compared to the operating voltage of the chip - usually between 7–20V)
- → electrons off the floating gate

J. Boukhobza, S. Rubini, Lab-STICC, Université de Bretagne Occidentale, 'Overview of Flash memory'

Flash memory: Write Operation



Program : FN tunneling

Off cell → Data "0"

Erase : FN tunneling

On cell → Data "1"

K. Choi, Samsung Electronics, co., Ltd
Flash design team 'Nand flash memory' 2010.

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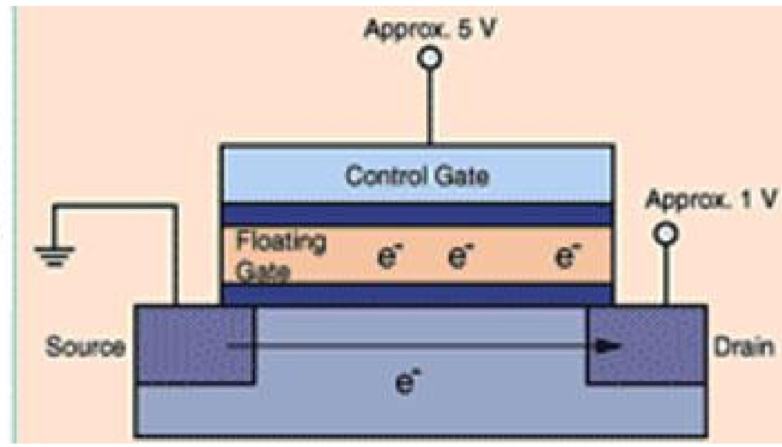
Flash memory: Read operation

Read

Applies the voltage at approx. 5 V to the control gate and at approx. 1 V to the drain. The state of the memory cell is distinguished by the current flowing between the drain and the source.



- If the selected cell has **low Vth** (logic ‘1’):
  - The transistor conducts → **current flows** through the NAND string.
- If the cell has **high Vth** (logic ‘0’):
  - The transistor does **not conduct** → **no current** flows.

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Flash memory:

- Faster in operation and more durable/reliable.
- Lighter and smaller in size.
- Uses less power and hence produces less heat
- Higher cost.
- Limited number of write and erase cycles

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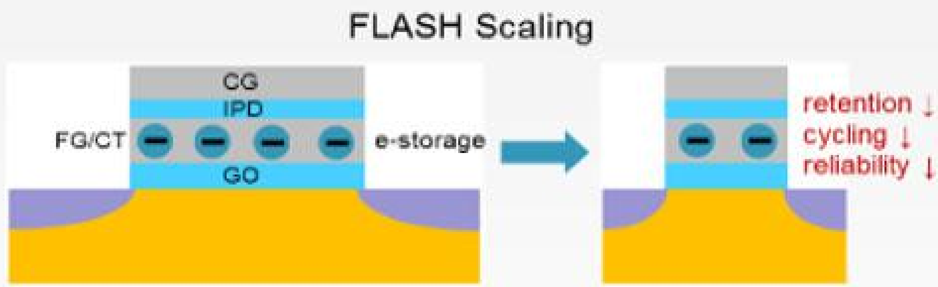
Flash memory-Challenges



- Information storage in Flash is based on charge density (C/cm^2)
- Losing a few electrons can cause severe reliability issues
- **Scaling causes exponentially increasing BER, reduced data retention and cycling**

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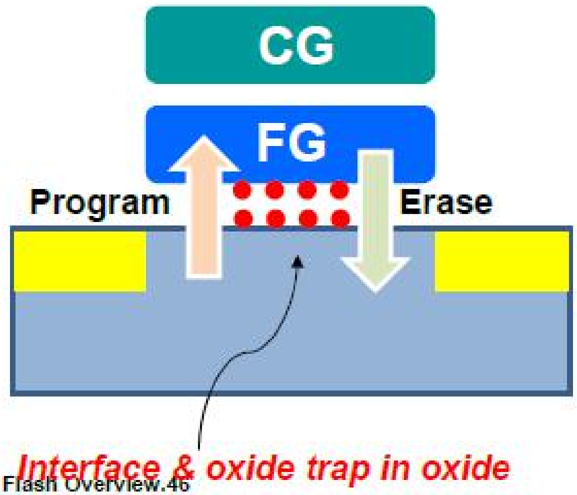
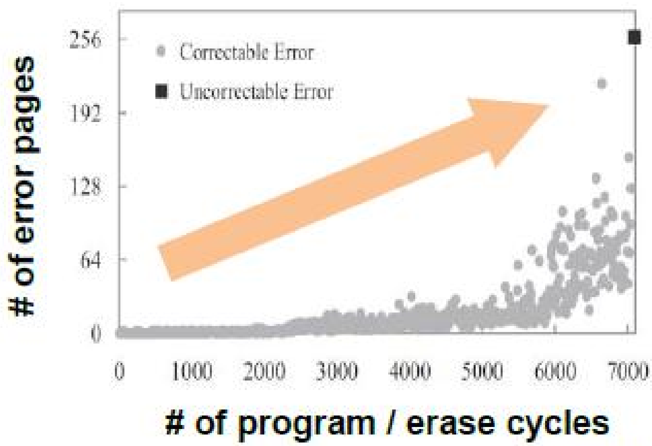
Flash memory-Challenges

- **Repeated program & erase operations wear-out NAND flash because of tunnel oxide degradation**
  - High voltage in program & erase operation give a damage to tunnel oxide, and then increase interface and oxide trap density
  - The larger number of P/E cycles, the higher BER (Bit error rate)
  - Tunnel oxide damage accelerates the reliability degradation, for example retention error and read disturb error



NAND Flash Overview.46



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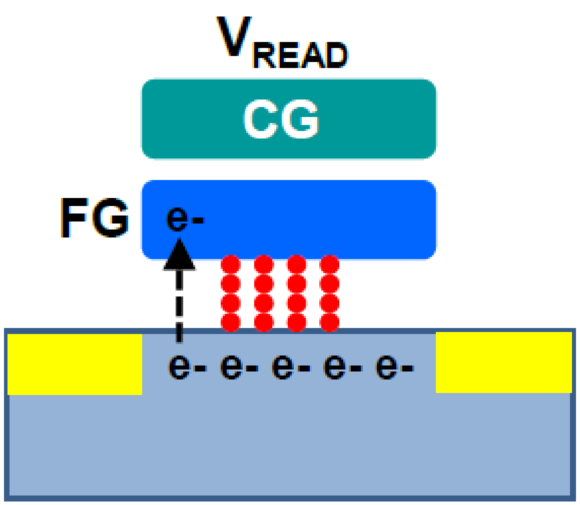
Flash memory-Challenges

- **Read disturb is an unexpected soft-program phenomenon due to repeated read operations**
  - V_{READ} (6~7V) which is applied to unselect W/L's in read operation can cause a soft-programming.
  - Read stress increases with the read voltage and read operation time
  - As P/E cycles increase, read disturb errors are accelerated



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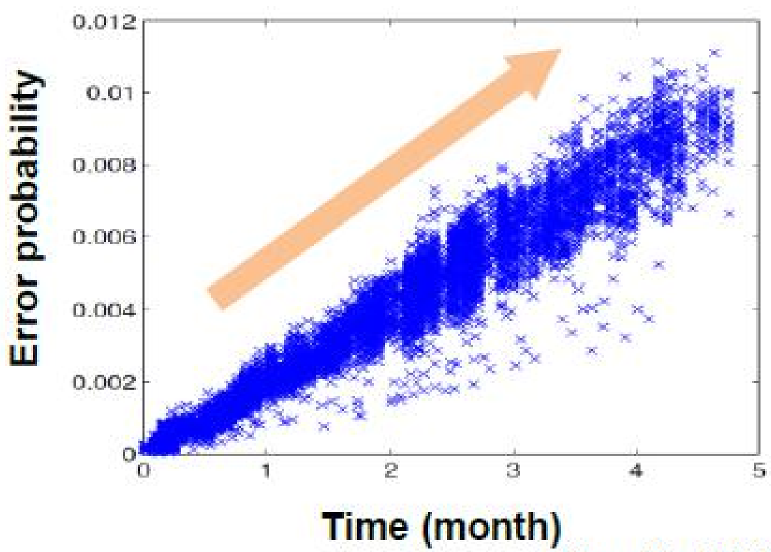
Flash memory-Challenges

- **The retention concept is the ability of a memory to keep a stored information over time with no biases applied**
  - Thermal stress (= high ambient temperature) leads to charge loss which means cell Vth shift to lower state
  - Charge loss causes an increase in BER, eventually read failures



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Flash memory-Challenges

Endurance

A limited number of program/erase operations

Retention

Data loss by leakage

Read Disturb

A limited number of read operations

Flash memory-Challenges

- Planar NAND is hitting physical limits at about the 1z nm node: The cell size cannot be reduced much further.
- 3D NAND relaxes the cell dimensions and adds additional layers in the vertical direction.
- 3D NAND appears to be the only viable option to increase densities until a better non-volatile mass storage technology emerges.

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Flash memory-Challenges

Comparison of Solid-State Memory Technologies (1)

	SRAM	DRAM	NOR Flash	NAND MLC Flash
Non-volatile	N	N	Y	Y
Memory cell factor (F^2)	50-120	6-10	10	4-5
Read time (ns)	1-100	30	10	50
Write/erase time (ns)	1-100	50	$10^5/10^7$	$10^6/10^5$
Number of rewrites	10^{16}	10^{16}	10^5	10^3
Power consumption at write	Low	Low	High	Med
Required input voltage (V)	None	2	6-8	1.8

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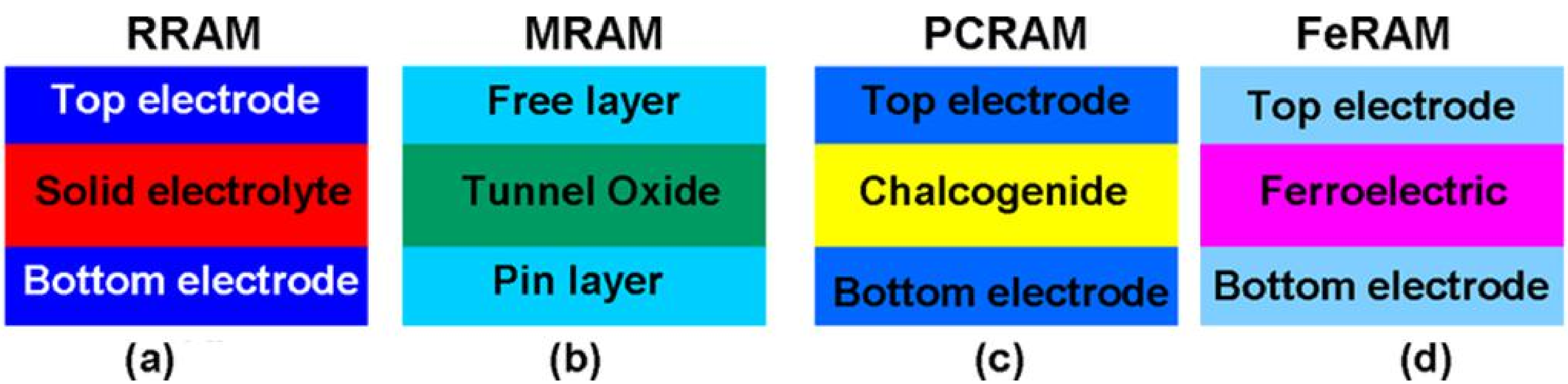
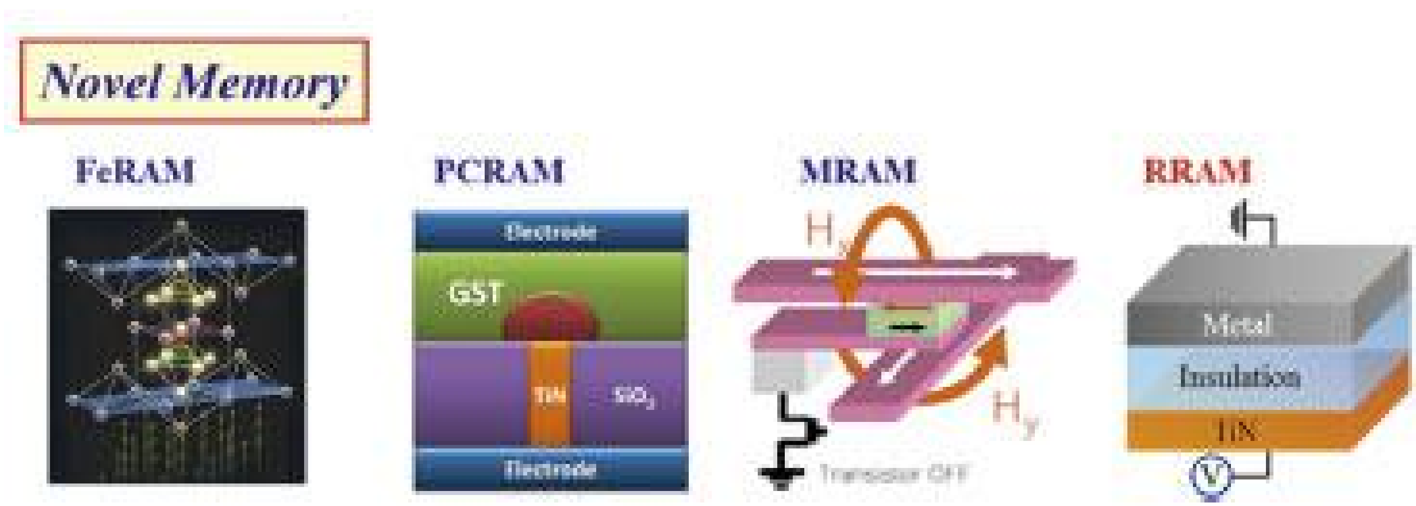
NV-Memory: Future





T.C. Chang, et al., “Resistance random access memory”, Materials Today (2016)

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